DEPARTMENT OF MECHANICAL & AEROSPACE ENGINEERING

University of Strathclyde Engineering

Icing Certification for Urban Air Mobility Systems under Appendix C Conditions

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Abstract

This project was conducted in collaboration with Airbus as part of its OSCAR programme. OSCAR takes up the position of a startup company based in the UK dedicated to designing, manufacturing and managing an Urban Air Mobility (UAM) aircraft. UAM is an emerging mode of transportation designed to reduce commute times and achieve sustainability goals. The UAM aircraft proposed and developed was a hybrid-electric Vertical Take-off and Landing (eVTOL) aircraft, powered by an electric generator coupled with a turboshaft engine. This paper provides guidance for achieving certification for Appendix C icing conditions under the EASA CS-23, SC-VTOL, CS-P and SC E-19 EHPS frameworks, tailored for an eVTOL aircraft. To aid in achieving certification, a certification compliance matrix was constructed. This involved deciphering certification regulations applicable to the OSCAR aircraft, extracting relevant icing requirements, and mapping each requirement to multiple testing methodologies, aimed at providing time and cost effective certification compliance strategies. A case study was performed to estimate the aerodynamic penalties caused by the most hazardous icing conditions within the Appendix C envelope. This was accomplished through adopting manual ice scaling methods to obtain an ice shape found in literature and utilising ANSYS CFD software to simulate flight behaviour. The case study demonstrated a 14% loss in Cl and a 169% increase in Cd at the cruise angle of attack of 4°, with a stall angle of 7° for the iced aerofoil. Furthermore, an extensive review of literature for anti-icing and de-icing technologies that include coatings, ice protection systems and ice detection was performed, highlighting suitable and effective technologies to be used in the OSCAR aircraft. These deliverables provide future OSCAR participants with the background and direction needed to progress towards achieving full Appendix C icing certification of the aircraft.

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Nomenclature

Symbol	Description	Units
Abbreviations and Acronyms		
AFM	Aircraft Flight Manual	—
AI	Artificial Intelligence	—
AMC	Acceptable Means of Compliance	—
ANSYS	Analysis System (commercial CFD software)	—
AoA	Angle of Attack	—
AR	Aspect Ratio	—
ASTM	American Society for Testing and Materials	—
CAA	Civil Aviation Authority (United Kingdom)	—
CFD	Computational Fluid Dynamics	—
CFR	Code of Federal Regulations (United States)	—
CFRP	Carbon Fibre Reinforced Polymer	—
CNT	Carbon Nanotubes	—
CS-23	Certification Specification for Normal Category Aeroplanes	—
CS-P	Certification Specification for Propellers	—
CSV	Comma-Separated Values	—
EASA	European Union Aviation Safety Agency	—
ED	EUROCAE Document	—
EHPS	Electric / Hybrid Propulsion System	—
EMEDS	Electromechanical Expulsion De-Icing System	—
EUROCAE	European Organisation for Civil Aviation Equipment	—
eVTOL	Electric Vertical Take-Off and Landing	—
FAA	Federal Aviation Administration	—
FENSAP-ICE	Finite Element Navier-Stokes Analysis Package for Ice (ANSYS icing CFD code)	—
FIKI	Flight Into Known Icing	—

Symbol	Description	Units
IAS	Indicated Airspeed	—
ICE-WIPS	Icephobic Coating and Electrothermal Wing Ice Protection System	—
ICTS	Ice Contaminated Tailplane Stall	—
IPS	Ice Protection System	—
IWT	Icing Wind Tunnel	—
JAXA	Japan Aerospace Exploration Agency	—
LEWICE	Lewis Ice Accretion (NASA icing CFD code)	—
LWC	Liquid Water Content	—
MAC	Mean Aerodynamic Chord	—
MED	Mean Effective Diameter	—
MOC	Means of Compliance	—
MOPS	Minimum Operational Performance Specifications	—
MRO	Maintenance, Repair, and Overhaul	—
MTOW	Maximum Take-Off Weight	—
MVD	Mean Volumetric Diameter	—
NACA	National Advisory Committee for Aeronautics	—
NASA	National Aeronautics and Space Administration	—
OSCAR	Airbus university collaboration programme (host of this project)	—
POSS	Polyhedral Oligomeric Silsesquioxane	—
RANS	Reynolds-Averaged Navier-Stokes	—
SA	Spalart-Allmaras (turbulence model)	—
SAE	Society of Automotive Engineers	—
SC E-19	Special Condition for Electric / Hybrid Propulsion Systems	—
SC-VTOL	Special Condition for Small-Category VTOL Aircraft	—
SLD	Supercooled Large Droplet	—
SLIPS	Slippery Liquid-Infused Porous Surfaces	—
SLW	Supercooled Liquid Water	—

Symbol	Description	Units
SPU	Slippery Polyurethane	—
SST	Shear Stress Transport (turbulence model)	—
TAS	True Airspeed	—
UAM	Urban Air Mobility	—
VTOL	Vertical Take-Off and Landing	—
WP	Work Package	—
Roman Symbols		
Ac	Accumulation parameter	—
b	Relative heat factor	—
c	Chord length	m
Cd	Coefficient of drag	—
Cl	Coefficient of lift	—
CL,max,clean	Maximum coefficient of lift, clean configuration	—
CL,max,iced	Maximum coefficient of lift, iced configuration	—
Cp	Coefficient of pressure	—
Cp,a	Specific heat capacity of air at constant pressure	J/(kg·K)
cp,ws	Specific heat capacity of water at the surface	J/(kg·K)
d	Characteristic dimension (2 × leading-edge radius)	m
Dv	Diffusivity of water vapour	m^2/s
g	Gravitational acceleration	m/s^2
h	Altitude	m, ft
H_C	Convective heat transfer coefficient	$W/(m^2·K)$
H_G	Gas-phase mass transfer coefficient	$kg/(m^2·s)$
K	Inertia parameter	—
K_0	Modified inertia parameter	—
K_{film}	Thermal conductivity of air at film temperature	W/(m·K)
L	Temperature lapse rate (troposphere)	K/m
L_f	Latent heat of freezing	J/kg

Symbol	Description	Units
n_0	Stagnation-point freezing fraction	—
Nu	Nusselt number at stagnation point	—
P_0	Sea-level standard pressure	Pa
P_{st}	Static pressure at altitude	Pa
P_{w_st}	Vapour pressure of water at static temperature	Pa
P_{w_surf}	Vapour pressure of water at surface temperature	Pa
Pr	Prandtl number at film temperature	—
r	Recovery factor	—
R	Specific gas constant of air	J/(kg·K)
Re_{MVD}	Droplet Reynolds number (based on MVD)	—
Re_{film}	Reynolds number evaluated at film temperature	—
RP	Range parameter	—
S	Sutherland constant	K
Sc	Schmidt number at film temperature	—
T_0	Sea-level standard temperature	K
T_{film}	Film temperature (mean boundary layer temperature)	K
T_{ref}	Reference temperature (Sutherland's law, 273.15 K)	K
T_{st}	Static (freestream) temperature at altitude	K, °C
T_{total}	Total (stagnation) temperature	K, °C
t_i	Leading-edge ice thickness	mm
V	Freestream velocity	m/s, kts
$V_{s,clean}$	Clean-configuration stall velocity	m/s, kts
$V_{s,iced}$	Iced-configuration stall velocity	m/s, kts
y^+	Dimensionless wall distance	—
Greek Symbols		
α	Angle of attack	°
β_0	Stagnation-point collection efficiency	—
ΔT_s	Surface temperature difference (Tref − 273.15 K)	K

Symbol	Description	Units
ΔT_{st}	Static temperature difference (Tst − 273.15 K)	K
λ/λ_{stokes}	Stokes ratio (droplet range correction)	—
Λ	Wing sweep angle	°
Λ_v	Latent heat of vaporisation	J/kg
μ_{air}	Dynamic viscosity of air at static temperature	Pa·s
$\mu_{a,film}$	Dynamic viscosity of air at film temperature	Pa·s
μ_{film}	Dynamic viscosity at film temperature	Pa·s
μ_{ref}	Reference viscosity at reference temperature	Pa·s
ρ_{air}	Air density at static temperature	kg/m³
ρ_{film}	Air density at film temperature	kg/m³
ρ_{ice}	Density of ice	kg/m³
ρ_{water}	Density of water	kg/m³
τ	Exposure time	s,min
ϕ	Water energy transfer parameter	—
θ	Air energy transfer parameter	—

1.0 Introduction

This section introduces the concept of Urban Air Mobility and eVTOL aircraft, outlines the hazards posed by in-flight icing and the role of certification when designing an eVTOL, and provides background on the Airbus OSCAR programme alongside the objectives of this project.

1.1 Urban Air Mobility & eVTOLs

Urban Air Mobility (UAM) is an emerging concept aimed at transforming the aviation industry through the integration of Electric Vertical Take-Off and Landing (eVTOL) aircraft as a mode of transportation. The adoption of eVTOL aircraft offers several benefits including shorter commute times, safer mobility, and progress in sustainability initiatives aimed at achieving net-zero targets. With the expansion of urbanisation, traditional modes of transportation such as roadways and public transportation systems are becoming increasingly inefficient for urban mobility. UAM offers a solution by introducing three-dimensional mobility, taking advantage of air travel to mitigate traffic congestion. Depending on the aircraft configuration, an eVTOL can complete a 30 km urban mission in 6.6 to 17.7 minutes [1]. As environmental issues such as climate change and unsustainable practices continue to intensify, the adoption of eVTOL technology has become imperative for decarbonising traditional transportation infrastructure. This is achieved through the substitution of fossil fuels to electric power, facilitating zero-emissions, alongside reduced noise pollution [2]. Therefore, eVTOLs represent a viable option for the advancement of the green economy.

There are several configurations for eVTOL aircraft, each suited for different operations, with the most notable being:

Wingless/Multirotor: This configuration prioritises efficiency in short-range urban missions. Fully relying on the rotors for lift and forward movement, the absence of a wing results in lower cruising speeds and shorter ranges. An example aircraft is the Volocopter VoloCity Air Taxi [1].

Lift + Cruise: Featuring a wing for efficient cruising, this configuration utilises two different propulsion systems, with one system being dedicated to vertical lift, and the other tailored to forward cruising. The incorporation of two different systems leads to an increase in weight, resulting in decreased speed. This configuration is featured on the Kitty Hawk Cora aircraft [1].

Vectored Thrust: Characterised by one propulsion system that manages both hovering and cruising through physical rotation of the thrust direction. This configuration is optimal for fast, long-range cruise flights; however, it requires complex control systems to facilitate the transition of modes safely. The Lilium Jet is an example of a vectored thrust eVTOL [1].

Over $1 billion has been invested into startups, with more than 130 concepts produced [1]. Major aircraft companies are already in the certification phase; however, the industry still has major challenges to overcome. Battery limitations remain a critical issue: high-performance vectored thrust aircraft impose significant power demands, which standard electric batteries are incapable of sustaining [1].

1.2 The Problem of Icing and the Role of Certification

Ground and in-flight icing are serious hazards to flight safety, with the aircraft experiencing severe performance degradation. Critical control systems that handle manoeuvrability and maintain safe operating conditions, such as the ailerons, elevators, rudders, windshields, pitot tubes, engine intakes, propellers and fuel nozzles can have their operation limited or, in some cases, be rendered unusable [3]. Inoperability of such critical components can drastically diminish the safety of the aircraft.

When ice forms on critical surfaces such as the wings, the profile of the aerofoil is altered, which can contribute to an increase in surface roughness [4]. This leads to a significant increase in the aerodynamic drag and a severe decrease in the maximum lift, which as a result can increase the risk of the aircraft stalling [5], making icing a contributing factor in numerous aircraft accidents [4]. The likelihood of experiencing flow separation also rises as by-product of ice accretion, potentially resulting in a complete loss of control of the aircraft. Ice shedding is another serious hazard; this hazard involves ice shedding from a surface such as the propeller and striking other aircraft components, potentially causing damage that may lead to a structural or a mechanical failure [4]. Furthermore, ice can accumulate and obstruct aircraft sensors, impairing their performance and increasing the possibility of inaccurate data delivery to the pilot [4].

eVTOL aircraft face exacerbated icing risks due to their distinct flight profiles and aircraft geometries. eVTOLs typically operate at lower altitudes with slower aircraft speeds, which increases exposure to high liquid water content environment, as well as a decrease in the effect of aerodynamic heating [4]. Smaller chord lengths, typically found on eVTOL aircraft, experience an increase in droplet collection efficiency, which can lead to more severe ice accretion [6]. Traditional IPS employed on commercial aircraft, such as engine bleed air heating systems, impose high power demands, and as eVTOL aircraft operate on electric batteries, such high-power demands would severely deplete the battery capacity. This leads to reduced flight time and range, thereby rendering most traditional protection systems impractical [7].

Given the severity of the hazards associated with aircraft icing, certification regulations and requirements have been established by aviation governing bodies such as the FAA and EASA, to ensure the aircraft is capable of operating within the requested flight envelope safely. Specific atmospheric icing conditions have been defined by the aviation authorities, such as Appendix C and Appendix O icing conditions [8]. This study focuses on Appendix C icing conditions, which are a collection of meteorological parameters consisting of variables such as altitude, temperature, liquid water content (LWC), and SLW droplet size that an aircraft must be able to withstand during operation. Appendix C is divided into two main conditions: Continuous Maximum conditions involving stratiform clouds and Intermittent Maximum conditions involving cumuliform clouds [8]. Compliance with the regulatory requirements can be demonstrated through providing sufficient testing data and evidence demonstrating that the aircraft and its respective systems are capable of surviving ice build-up and fulfilling the demands of the specified requirement.

1.3 The Airbus OSCAR programme and the Project Objective

This project was conducted in conjunction with Airbus as part of the OSCAR programme. The OSCAR programme tasks fourth and fifth year Mechanical and Aero-Mechanical Engineering students to assume the position of a newly formed startup company, which is involved in the development and manufacturing of a UAM/eVTOL aircraft. The company is based in the United Kingdom and therefore, falls under CAA guidelines. The OSCAR programme divides its participants into five distinct Work Packages (WPs), each tasked with a different objective.

WP0 is responsible for project management and oversees the business and marketing decisions, ensuring the design concept is feasible based on the available budget. WP1 is the design team, tasked with the design of the entire aircraft. This includes the structure, powerplant, engine, propeller, interior configuration, etc. WP2 is the manufacturing team, tasked with evaluating the concepts designed by WP1, and assessing their feasibility. This team also designs and organises the manufacturing facilities. WP3 is the services and operation team, which includes the management of the fleet, pilot training, MRO, and service modelling. WP4 is responsible for certification and maintenance, and is the WP of this project, which is the certification of the OSCAR aircraft for Appendix C icing conditions.

Although each goal within each WP constituted an individual project, the OSCAR programme required communication and coordination between all the WPs, due to the interlinking nature. Design decisions made by WP1 directly influenced the certification requirements analysed in WP4, and business decisions of WP0 impacted WP4, as the region choice for aircraft deployment determined the aviation authority and certification criteria.

Each WP was assigned an Airbus supervisor and an academic supervisor, tasked with overseeing the project and aiding in the development of deliverables. Monthly meetings were held across both semesters, and involved the construction of progress reports, and the determination of interdependencies between each WP.

The primary objective of this project was the development of a certification matrix for a hybrid-electric eVTOL aircraft, operating in Appendix C icing conditions, ensuring testing methodologies were mapped to their respective requirements. The CAA has adopted the EASA technical requirements; therefore, the EASA frameworks were thoroughly examined. The certification frameworks applicable to the OSCAR aircraft and examined in this project were CS-23 Amendment 6 Issue 4, SC-VTOL Issue 2, CS-P Amendment 2, and SC E-19 Issue 1.

Further objectives include the evaluation of anti-icing and de-icing technologies and their applicability to the OSCAR aircraft, as well as a case study to estimate the aerodynamic degradation experienced by the OSCAR aircraft due to critical ice formation (Appendix C conditions). The results of the case study were relayed to the design team, allowing for the aircraft to be designed with these penalties accounted for, whilst also satisfying the certification requirements.

2.0 Literature Review

This section outlines the physics governing atmospheric ice accretion, specifically Appendix C icing. The regulatory history and structure of the Appendix C icing certification framework was reviewed. A survey on active and passive ice protection technologies applicable to eVTOL aircraft was conducted, and discusses the scaling methodology used to estimate aerodynamic degradation caused by icing.

2.1 Physics of Atmospheric Icing

In-flight icing occurs when an aircraft is exposed to Supercooled Liquid Water (SLW) droplets; these droplets can remain in the liquid phase at temperatures as low as -40°C [3]. SLW droplets exist in a metastable state, as the absence of a seed particle means they are unable to undergo crystallisation and thus, retain their liquid state [9]. When a droplet makes contact with the airframe, the surface acts as a catalyst for heterogeneous nucleation. The kinetic energy of the droplet at impact disrupts its metastability, initiating a rapid phase transition that facilitates the formation of ice on the aircraft [9].

There are two partially overlapping stages that constitute the freezing process for droplets. The first stage is the recalescence freezing stage; this involves the release of latent heat, resulting in a sudden increase in temperature back to its standard freezing point. An ice crystal scaffold is created after this stage [9]. The second stage is the isothermal stage. Occurring immediately after the creation of the ice crystal scaffold, this stage involves the solidification of the remaining liquid water. The leftover liquid water freezes, growing radially inward and occupying the empty space inside the ice crystal scaffold formed during the first stage [9]. Latent heat is dissipated through conduction between the droplet and the contact surface, completing the freezing process and forming adhesive ice [9].

Ice accretion morphology is governed by the balance between aerodynamic heating and convective cooling at the surface, together with the impacting droplets mean effective diameter (MED), or equivalently the mean volumetric diameter (MVD) used in modern icing standards [9]. The proportion of aerodynamic heating and convective cooling experienced by a droplet determines whether the droplet will freeze instantly or flow rearward towards the trailing edge, where it may subsequently undergo accretion as ice [3].

Aerodynamic heating transpires through the conversion of the kinetic energy of the airstream to thermal energy [9]. Although aircraft icing becomes negligible when travelling at speeds greater than 575 knots [10], UAM aircraft such as eVTOLs typically operate at much slower speeds, rendering them vulnerable to aircraft icing. Convective cooling occurs through the loss of thermal energy from the aircraft surface by the cold ambient airstream [9].

Ultimately, the net thermodynamic balance between the aerodynamic heating and convective cooling determines the classification of the ice growth regime of the impacting droplets. A dry growth regime occurs when the convective heat losses exceed the aerodynamic heat gains, causing droplets to freeze completely upon impingement. This results in the formation of rime ice, which is characterised by its milky and opaque appearance, and low formation temperatures (-15°C) within low water content (LWC) environments [3].

As a result of the instantaneous freezing of the droplets during the recalescence stage, the formation of runback ice is improbable. Therefore, rime ice grows forward against the direction of the airflow, forming rough protrusions. While this produces an increase in drag and decrease in lift, the aerofoil profile does not experience a substantial risk of flow separation which would warrant a sudden stall [11].

Conversely, a wet growth regime occurs when aerodynamic heat gains exceed convective heat losses. The latent heat released during the initial freezing maintains the local surface temperature at 0°C [9]. Droplets partially freeze upon impact; the leftover liquid droplets are blown off by the oncoming airflow, resulting in runback ice and severe aerodynamic penalties [3]. Air bubbles are allowed to escape due to the freezing stage occurring more slowly, making the ice more dense and tougher. This type of ice is referred to as glaze ice [3].

Glaze ice is formed within the wet growth regime, characterised through the appearance of horn-like shapes caused by water freezing at the upper and lower points of the leading edge. These horns are perpendicular to the airflow and act as flow disruptors, significantly increasing drag, reducing lift, and inducing flow separation [11]. The aircraft becomes prone to stalling at standard operating speeds and angles of attack [11]. During extreme glaze icing conditions, an unprotected aircraft can experience up to a 200% increase in the coefficient of drag [12] coupled with a reduction in the maximum lift by 25-30% [13]. Such factors correspond to a significantly higher stall speed.

Based on these characteristics, glaze ice was considered the most critical icing morphology [12]. Glaze icing threats are further amplified for eVTOL aircraft due to their unique profiles and structural sensitivities. This translates to an increased rate of ice accumulation arising from smaller chord lengths, and asymmetric shedding of the rotors which can lead to ice ingestion hazards [3]. Unregulated ice accretion along the airframe contributes to irregular pressure distributions, altering the behaviour of critical control surfaces, potentially resulting in uncontrollable piloting. The accumulation of ice can cause a partial or complete blockage of pitot tubes and static vents, inducing misleading flight measurements, producing incorrect airspeed and altitude data.

2.2 Certification Framework for Icing

The Appendix C icing envelope was developed in the 1940s and 1950s by researchers from the U.S. Weather Bureau and National Advisory Committee for Aeronautics (NACA) [14]. The Appendix C envelope was constructed through NACA flight tests measuring and collecting U.S. continental atmospheric conditions employing rotating multi-cylinders to determine mean effective diameter (MED) [14],[15].

The Civil Air Regulations formally introduced the Appendix C icing envelope in 1955 with the release of Amendment 4b-2 [15], defining cloud extents in statute miles, and ranges for mean effective diameter and liquid water content [15]. These regulations were later reclassified into 14 CFR Part 25 in 1965, later being formally adopted by EASA in 1993 [15],[16]. OSCAR is a startup based within the UK; its type-certification applications therefore fall under the UK Civil Aviation Authority (CAA). Following Brexit, the CAA has adopted the EASA certification specifications along with any subsequent amendments as of 31 December 2020 [17],[18]. Therefore, the EASA certification specifications and regulations were followed, with the distinction that applications for type certification would be submitted to the CAA rather than EASA.

The EASA CS-23 Appendix C icing envelope comprises two atmospheric icing categories: Continuous maximum and intermittent maximum [8]. Continuous maximum conditions are characterised by prolonged, lower intensity icing associated with stratiform clouds [8],[19]. Droplet characteristics for this type of condition involve a moderate liquid water content (LWC), between 0.2 and 0.8 g/m^3, with a mean effective droplet diameter (MED) between 15 and 40 µm, and ambient temperatures ranging from 0°C to -30°C [8]. The standard horizontal extent for continuous maximum conditions defined by the regulations is 17.4 nautical miles [8]. This type of icing condition is relevant during cruise and holding phases of flight, where the aircraft may be immersed in stratiform clouds for extended periods of time.

Intermittent maximum conditions are defined by short-duration, high-intensity icing associated with cumuliform clouds. Cumuliform clouds entail powerful updrafts; these conditions support the development of denser supercooled water droplets when compared to stratiform clouds. Droplet characteristics for intermittent maximum icing conditions involve a high liquid water content (LWC), between 0.25 and 2.9 g/m^3, with a similar mean effective droplet diameter (MED) to the continuous maximum conditions, ranging between 15 and 50 µm. Temperature within cumuliform clouds vary between 0°C and -40°C [8],[19]. The standard horizontal extent for intermittent maximum conditions defined by the regulations is 2.6 nautical miles [19].

A correction factor, defined as the "F-factor" is introduced in the regulations; this adjusts the liquid water content based on the horizontal extent of the cloud encountered [15]. The purpose of defining a correction factor is to assist manufacturers and applicants intending to design an ice protection system capable of withstanding Appendix C icing conditions. If an engineer needs to estimate the effects of continuous maximum icing on an unprotected aircraft surface over a distance of 100 nautical miles, the correction factor allows them to obtain an adjusted liquid water content value that is accurate for the scenario they wish to assess [15].

Intermittent maximum icing is considered more hazardous than continuous maximum icing, in part due to possessing a higher water mass flux. Cumuliform clouds are characterised by their high liquid water content coupled with warmer temperatures; this combination leads to a low freezing fraction as the latent heat of fusion dissipates more slowly. Glaze ice formed within intermittent maximum conditions generates further protruding horn shapes when compared to glaze ice in continuous maximum conditions. The increased length of the protruding horns results in greater flow separation, leading to a severe loss in the maximum lift while inducing immense amounts of drag, making the aircraft difficult to pilot [15].

It is important to note that the Appendix C envelope does not encompass all atmospheric icing threats. Supercooled large droplets (SLD) conditions involve droplets possessing a greater mean effective diameter and take into account splashback caused by large droplets breaking into smaller droplets upon impact. However, this project focuses exclusively on Appendix C conditions; therefore, any icing threats related to SLD conditions will not be taken into consideration.

The CS-23 framework provides the foundational airworthiness requirements for small normal-category aircraft. This contains a defined Appendix C envelope with graphs and data that can be deciphered and utilised when testing for compliance. CS 23.2165 requires the applicant to demonstrate safe operation within the requested icing conditions [20]. CS 23.2415 mandates that the aircraft and powerplant installation design must be capable of preventing critical ice accumulation that would adversely affect the engine [20]. CS 23.2540 states that the IPS must be capable of protecting the aircraft and ensuring safe operation when operating under FIKI conditions [20]. However, the CS-23 framework was designed to serve aircraft with conventional fixed wings and it does not encompass the unique attributes of a VTOL aircraft such as the reliance on electric batteries.

To bridge this gap, for VTOL applications, EASA published SC-VTOL Issue 2, which is a special condition designed for small category VTOL-capable aircraft. This introduced several additional requirements for icing. VTOL.2165 requires that the aircraft be capable of safe operation within the requested envelope, mirroring the requirement CS 23.2165 [21]. VTOL.2415 requires adequate ice protection for lift/thrust systems, while VTOL.2430 emphasises the importance of energy reserve management [21]. Propeller specific icing requirements are addressed under CS-P Amendment 2; CS-P 350c, CS-P 380, and CS-P 390 require the aircraft propeller be equipped with de-icing systems capable of withstanding centrifugal loads, lightning strikes, and surviving endurance testing [22]. SC E-19 Issue 1 was released to supplement existing certification for electric and hybrid propulsion system aircraft [23]. CS-23, SC-VTOL, and SC E-19 contain overlapping requirements, which highlights the absence of a consolidated compliance framework for eVTOLs, as applicants are required to cross-reference multiple certification documents to construct a complete certification matrix.

The primary means of compliance document used for icing certification is ASTM F3120/F3120M-20, which specifies ice protection system performance, flight procedures, and instrumentation requirements [19]. EUROCAE published ED-314 in 2024, addressing VTOL specific compliance methodologies for encounters involving inadvertent icing encounters [24]. Currently, no single consolidated framework exists that outlines these requirements with their respective compliance methodologies for eVTOL platforms.

2.3 Active and Passive Ice Protection Technologies

The selection or design of an appropriate ice protection system (IPS) is essential for ensuring the aircraft remains protected against icing threats. Implementing an effective IPS strengthens the case for compliance presented to the certification authority, therefore demonstrating airworthiness under the requested icing conditions.

Traditional IPS equipped on commercial aircraft, such as engine bleed-air heating or pneumatic de-icing boots provide efficient anti/de-icing capabilities. However, these traditional IPS would not retain their efficiency if equipped on eVTOL aircraft. This is because they impose high power demands and weight penalties which are incompatible with eVTOL energy budgets, as the propulsion system relies on electric batteries or a hybrid-electric generator with limited power reserves [7]. SC-VTOL.2430 mandates that the energy storage system must provide enough energy for "sufficient reserve based on standard flight", emphasising the need for energy efficient solutions. This constraint renders many traditional protection systems impractical for VTOL usage, prompting researchers to investigate alternative technologies.

Active ice protection systems can either operate for anti-icing or de-icing purposes. As active systems require external energy, ensuring the selection of an energy efficient system is essential, especially for eVTOL applications.

Electro-thermal heating systems are the most widely researched IPS for eVTOLs. These systems utilise electrical energy to generate heat. Heating elements are typically either applied as thin surface coatings or are embedded into the surface [25]. Electro-thermal heating systems operate through a mechanism referred to as Joule heating [26]. What separates electro-thermal heating systems from conventional hot-air systems is the energy efficiency. The heaters can be powered off in minimal icing conditions, allowing the pilot to maximise energy efficiency [25]. Furthermore, these systems are lightweight and are easy to implement, making them highly suited for the leading edge of the wing and propellers [27]. There are concerns when implementing this technology on non-metallic surfaces, such as CFRP. This system offers poor lateral heat conduction when applied on CFRP surfaces, resulting in the appearance of cold gaps which can lead to the formation of runback ice [28].

Carbon nanotubes (CNT) are a low weight alternative to other forms of active IPS. Characterised with a density of 0.019 g/m^2, CNTs are manufactured into thin webs and are integrated into the fabrication of the aircraft surface [29]. This system operates under the same joule heating principle as the electro-thermal heating system [29]. The heating profile of this system can be configured by altering the orientation of the CNT webs during installation, making this system highly configurable [29]. These qualities make CNTs an attractive option for eVTOL application. Furthermore, there are no conflicts between the certification requirements and the mechanisms of CNT [19]. However, manufacturing and implementation are complex due to their hypersensitive nature; misjudgements in applied pressure can alter the system performance, potentially compromising performance [29].

Electromechanical expulsion de-icing systems (EMEDS) are another option when considering active IPS. This system utilises electromagnetic coils or ultrasonic wave generators to fracture and shed ice formed [30]. Mechanical actuators are installed inside or underneath the skin of the wing or rotor. When ice forms, the actuators generate a mechanical shockwave/excessive vibration that travels outward through the composite skin, resulting in a reduction of the ice adhesive strength, allowing the airstream to expel the ice. Cox & Co has achieved FAA certification for FIKI on Part 23 aircraft, making this proven for CS-23 aircraft [31]. However, extensive trajectory and component analysis would need to be conducted to ensure shed ice will not damage critical components. VTOL.2415 and ASTM F3120/F3120M-20 both require extensive analysis and testing of ice shedding hazards [19],[21].

Passive coating systems are key in minimising energy consumption when operating under icing envelopes. Since they are applied pre-flight either during manufacturing or maintenance routines, they operate without the use of external energy, making them highly suited for eVTOL applications. These systems modify the surface properties of the aircraft skin to reduce the ice adhesion strength. This allows the active icing system to easily melt the ice while minimising its energy expenditure, or letting the incoming airstream blow it off the wing. Passive coatings can be broadly categorised into two mechanisms: slippery liquid-infused porous surfaces (SLIPS) and low surface energy solid coatings.

Slippery Liquid-Infused Porous Surfaces (SLIPS) represent a modern approach to passive systems. SLIPS was developed with the intention of overcoming the durability weaknesses of superhydrophobic surfaces. A SLIPS coating is implemented into an aircraft by creating a micro or nanoporous solid substrate infused with a low surface tension liquid lubricant, such as silicone or fluorinated oil [32]. Rather than trapping air like traditional superhydrophobic coatings, the porous structure behaves as a sponge which locks the lubricant in place through van der Waals forces and hydrogen bonding [33]. The end product is a highly stable, smooth, slippery surface which is completely immiscible with incoming water, preventing ice from anchoring onto the aircraft [33].

Under glaze icing conditions, aircraft equipped with SLIPS coatings experience ice weight reduction up to 64% when compared to uncoated surfaces [34], as well as ice adhesive strength values below 9 kPa [35]. This allows SLIPS to shed ice naturally, without the assistance of active systems to weaken the ice.

Low surface energy coatings combat icing by minimising the molecular interactions between the droplet and the aircraft surface. They employ polymers such as fluoropolymers and polysiloxanes to reduce the van der Waals forces and hydrogen bonding at the surface. This weakens the adhesive strength of the ice, facilitating easy removal [36]. When tested under glaze icing conditions, modified polyurethane topcoats incorporating nanosilica and polyhedral oligomeric silsesquioxane (POSS) experienced a 65% reduction of ice accretion when compared to unmodified surfaces [37].

However, despite these results, passive coatings alone are insufficient as they offer little resistance to rain erosion [38],[39]. Active systems require high amounts of energy, which is unsustainable for eVTOL aircraft with their limited power supply. The limitations of active and passive systems emphasised the necessity of developing a low-power, high-efficiency IPS. This requirement brought about the development of hybrid systems, which are characterised by the supplementation of passive coating systems with an active heating system.

An efficient hybrid system examined was the Icephobic Coating and Electrothermal Wing Ice Protection System (ICE-WIPS) architecture, which was developed by JAXA and NASA, involves installing a small electro-thermal heater at the leading edge of the wing that runs intermittently, using just enough power to melt the ice at the contact surface [38]. An icephobic coating was applied directly behind the heated area, which swept away the partially melted ice with the assistance of the incoming airflow. Wind tunnel tests proved the system was effective with a 70% reduction in energy consumption [38].

Certification requirements demand the installation and operation of an ice detection system. ASTM F3120/F3120M-20 outlines requirements for primary and advisory ice detection systems, mandating compliance with EUROCAE ED-103A or SAE AS5498A, which prioritise response time [19].

For ice detection, fibre optic sensors provide the highest measurement accuracy, capable of determining ice thickness, up to 10 mm, and distinguishing between ice morphologies [40]. Performance degradation prediction software was found to be the most modular solution; it compared real-time aerodynamic data to baseline models [41]. This approach incurs no weight penalties and offers minimal cost while satisfying ASTM F3120/F3120M-20 Annex A4 requirements [41].

2.4 Case Study: Icing Scaling and Aerodynamic Degradation

Estimating the aerodynamic penalties without the use of specialised icing CFD software, such as ANSYS FENSAP-ICE or NASA LEWICE, requires analytical scaling methods based on dimensionless parameters governing droplet impingement and freezing behaviour. Scaling methods were developed to allow researchers to perform ice wind tunnel tests on a subscale model experiencing the same ice accretion characteristics that a full scale model would experience, minimising testing costs.

Langmuir and Blodgett derived mathematical relationships between droplet trajectory and aerofoil geometry, leading to the discovery of the modified inertia parameter, K_0. The modified inertia parameter represents the ratio of droplet inertia to aerodynamic drag. This ratio was determined by the droplet diameter, freestream velocity, air viscosity and a characteristic body dimension [42]. Langmuir and Blodgett further established that if the modified inertia parameter fell below 1/8, then no droplet impingement would occur, defining a critical threshold for icing onset [42].

The stagnation point collection efficiency, β_0, defined as the fraction of freestream water that struck the aircraft surface was derived using the modified inertia parameter, and is validated against data predicted by the LEWICE code [43]. An acceptable tolerance for collection efficiency when selecting an ice shape to scale is $\pm 10\%$ [44].

Anderson expanded upon the framework by introducing two additional non-dimensional similarity parameters, the accumulation parameter and the freezing fraction [45]. The accumulation parameter, A_c, quantifies the total mass of ice deposited per unit area over the exposure duration; it is expressed as the product of LWC, freestream velocity, collection efficiency, and exposure time, normalised by the ice density and characteristic length [45]. The freezing fraction, n, derived from the

Messinger energy balance model, represents the proportion of water droplets that froze at the point of impact [45]. A freezing fraction value of unity represents rime ice conditions, whereas a freezing fraction value of less than unity represents glaze ice conditions [45].

When verifying an ice shape to be used for scaling, the freezing fraction between the scaled and reference icing scenario should be within 10% and the accumulation parameter within 15% [44]. The freezing fraction and accumulation parameter are considered the most influential parameters when conducting an ice scaling study [44]. Additionally, Anderson advises a stagnation temperature no warmer than -3°C to maximise reliability of the results [44]. Modifications to the scaling methods to facilitate swept-wing geometries were developed, incorporating the sweep angle into the collection efficiency and the convective heat transfer coefficient calculations [46]. There is currently no documented application of these scaling methods to estimate aerodynamic degradation for an eVTOL aircraft operating under the Appendix C envelope.

3.0 Methodology

Thus section details the methodology employed when constructing the certification compliance matrix, the ice shape scaling similarity parameters and aerodynamic degradation case study, and the comparative evaluation of anti-icing and de-icing technologies.

3.1 Certification Compliance Matrix

A comprehensive certification compliance matrix was constructed to address the absence of a consolidated framework for icing certification that maps requirements to testing methodologies, and is applicable to an eVTOL platform. The certification compliance matrix was developed through a systematic analysis of the required regulatory frameworks: EASA CS-23 Amendment 6, SC-VTOL Issue 2, CS-P Amendment 2, and SC E-19 Issue 1. These four frameworks were selected for their applicability to eVTOLs, specifically the OSCAR aircraft. The development process is illustrated in **Figure 1**.

Each subsection within the regulatory documents was examined. This was to identify and evaluate icing related requirements. A requirement was deemed icing relevant if it pertained to aircraft performance in icing conditions, crew visibility, propeller or rotor ice shedding, energy management under icing conditions, limitations to be recorded in the aircraft flight manual, and IPS design or installation requirements. Additionally, requirements addressing broader environmental conditions, such as lightning strikes, were included within the certification matrix, as a clear connection to icing was determined. Requirements possessing no conceivable relevance to icing were excluded from the framework; this prevented the matrix from being diluted by unrelated non-icing requirements. Furthermore, another WP4 member was tasked with the construction of a general certification matrix. To prevent unnecessary overlap between projects, and with the scope of this project being defined specifically around icing certification for Appendix C conditions, general certification requirements were excluded.

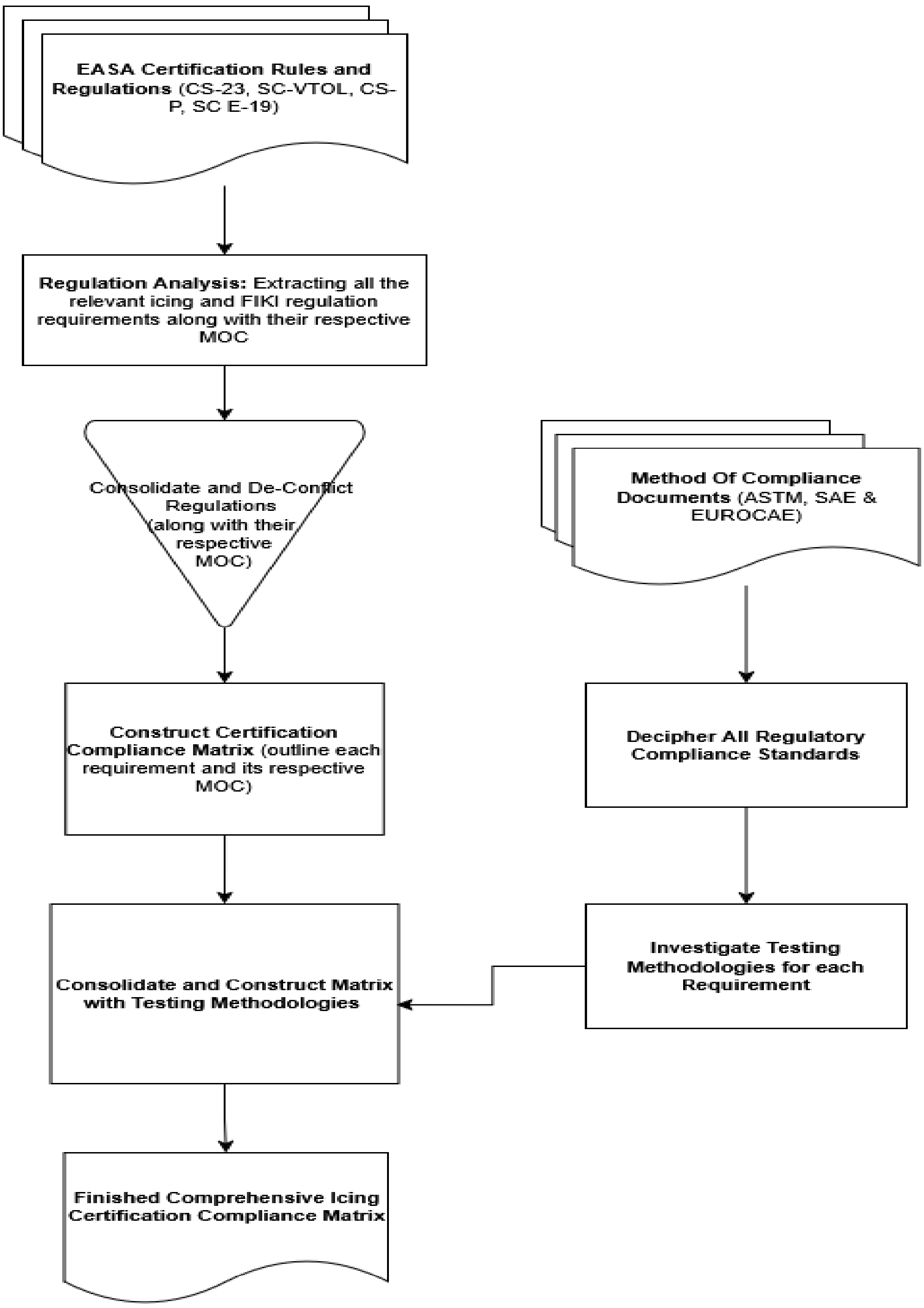

Figure 1: Certification Compliance and Testing Matrix Flowchart

Overlapping and conflicting requirements across certification frameworks were consolidated and filtered, along with their respective means of compliance. For example, requirements for IPS installed on the propellers are outlined by CS 23.2415, VTOL.2415, and CS-P 350c, with each requirement illustrating a different regulatory perspective. Overlapping requirements were consolidated with the differing perspective noted. Conflicting requirements were thoroughly examined, ensuring discrepancies were managed; this involved adopting the most conservative requirement between conflicting requirements and noting the discrepancies if deemed necessary.

This process was essential to maximise testing efficiency, preventing identical requirements from undergoing multiple testing campaigns, and facilitating cost and time effective strategies. The consolidated requirements were recorded in the comprehensive certification compliance matrix, with their respective methods of compliance.

The format of the certification compliance matrix is illustrated in **Figure 2.** Each entry records the certification specification along with its respective version, ensuring that updates made by the certification authority can be input efficiently. The certification requirement ID (e.g. CS 23.2415) and a description of the requirement were noted. A method of compliance (MOC) matrix was linked to the respective requirement, showcasing all the requirements within the MOC standard with multiple testing methodologies listed respectively. Estimates in testing time and costs were added if applicable and attainable. As OSCAR is an unknown startup company, obtaining costs and time estimates became unrealistic due to the lack of project maturity within the manufacturing and management teams. These fields were left as placeholders for future OSCAR candidates to complete as the project matures.

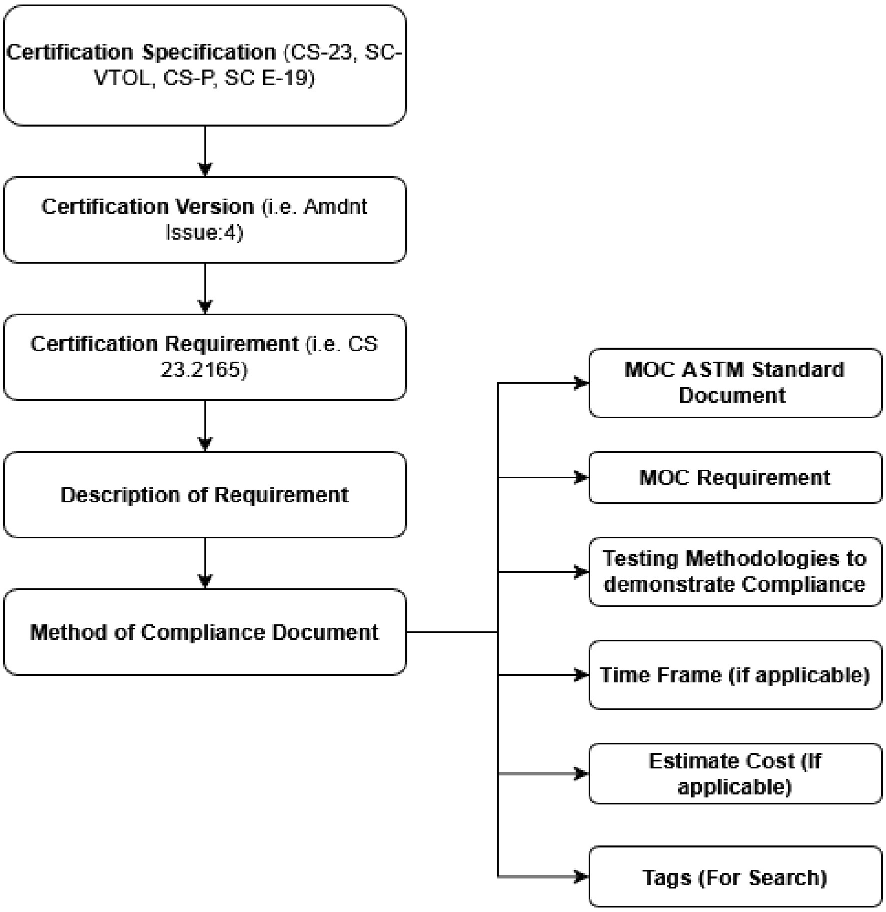

Figure 2: Format of Certification Compliance Matrix

The primary MOC documents analysed were ASTM F3120/F3120M-20, EUROCAE ED-314, ASTM F3316/F3316M-19 and ASTM F3066/F3066M-18. These documents were explicitly stated as acceptable means of compliance documents for certain requirements. For example, CS 23.2165 lists ASTM F3120/F3120M-20 as an acceptable means of compliance standard, meaning if all the requirements listed within

ASTM F3120/F3120M-20 are satisfied with sufficient testing data collected and evidence documented, then the certification authority would consider the requirement, CS 23.2165, as fulfilled.

At the time of this work, a newer revision of ASTM F3120/F3120M was available; however, the decision to remain with the -20 revision was made, as this version remains the primary means of compliance document accepted for icing certification by EASA.

Each MOC document was systematically broken down into subsections containing individual compliance requirements. Multiple testing methodologies were investigated for each requirement and documented within the matrix. Testing methods were derived from what the MOC documents deemed acceptable. The three main categories of testing methods were: computational analysis using software such as ANSYS FENSAP-ICE, icing wind tunnel (IWT) testing, and natural icing flight testing.

Natural icing flight testing offers the highest fidelity; however, it suffers from high testing costs. IWT testing involves 3D printing critical ice shapes and attaching them to the aircraft to observe aerodynamic characteristics when simulated under the applicant's chosen conditions. IWT allows the user to manipulate atmospheric and icing variables, such as the temperature, LWC and MVD, at a lower cost than the natural flight testing. Computational analysis incurs the lowest cost while also producing highly reliable results. As OSCAR is a startup, minimising costs is vital. This made computational analysis testing the most applicable testing methodology, as this yields highly accurate results while incurring the lowest costs among the discussed methodologies.

Consolidating multiple testing methods enabled cost and time effective certification strategies to be developed, allowing several requirements to be tested within a single testing campaign. The completed certification compliance matrix serves as the primary deliverable for this project as part of WP4 of the Airbus OSCAR programme.

3.2 Case Study: Ice Shape Scaling and Aerodynamic Degradation

Since the design team had yet to select an IPS, this case study aimed to estimate the aerodynamic penalties caused by the worst-case ice accretion on an unprotected NACA 0012 wing. The performance degradation data was provided to the design team, allowing them to build upon the estimates to facilitate safe operation, even in critical icing conditions. ASTM F3120/F3120M-20 defines the "worst-case" icing as a "22.5 min hold at minimum holding airspeed plus 10 knots" at the most critical icing conditions within the standard horizontal cloud extent [19].

Key wing and aircraft parameters used for the case study were supplied by the OSCAR WP1 design team, shown in **Table 1**:

Table 1: Wing and Aircraft parameters supplied by WP1 (Design Team)

Parameter	**Value**
Aerofoil	NACA 0012
Mean Aerodynamic Chord	2.27734m
Wingspan	17 m
Aspect Ratio	8.5
Wing Type	Conventional
Wing Mass	400 kg
Tailplane Mass	80 kg
Sweep Angle on Wings (°)	18.3274
Primary Material:	CFRP
Holding Airspeed (IAS)	190 kts
Analysis altitude	10000 ft
True Airspeed at 10000ft	102.88 m/s

A flow chart showcasing the logic throughout the case study was developed; this is illustrated in **Figure 3**.

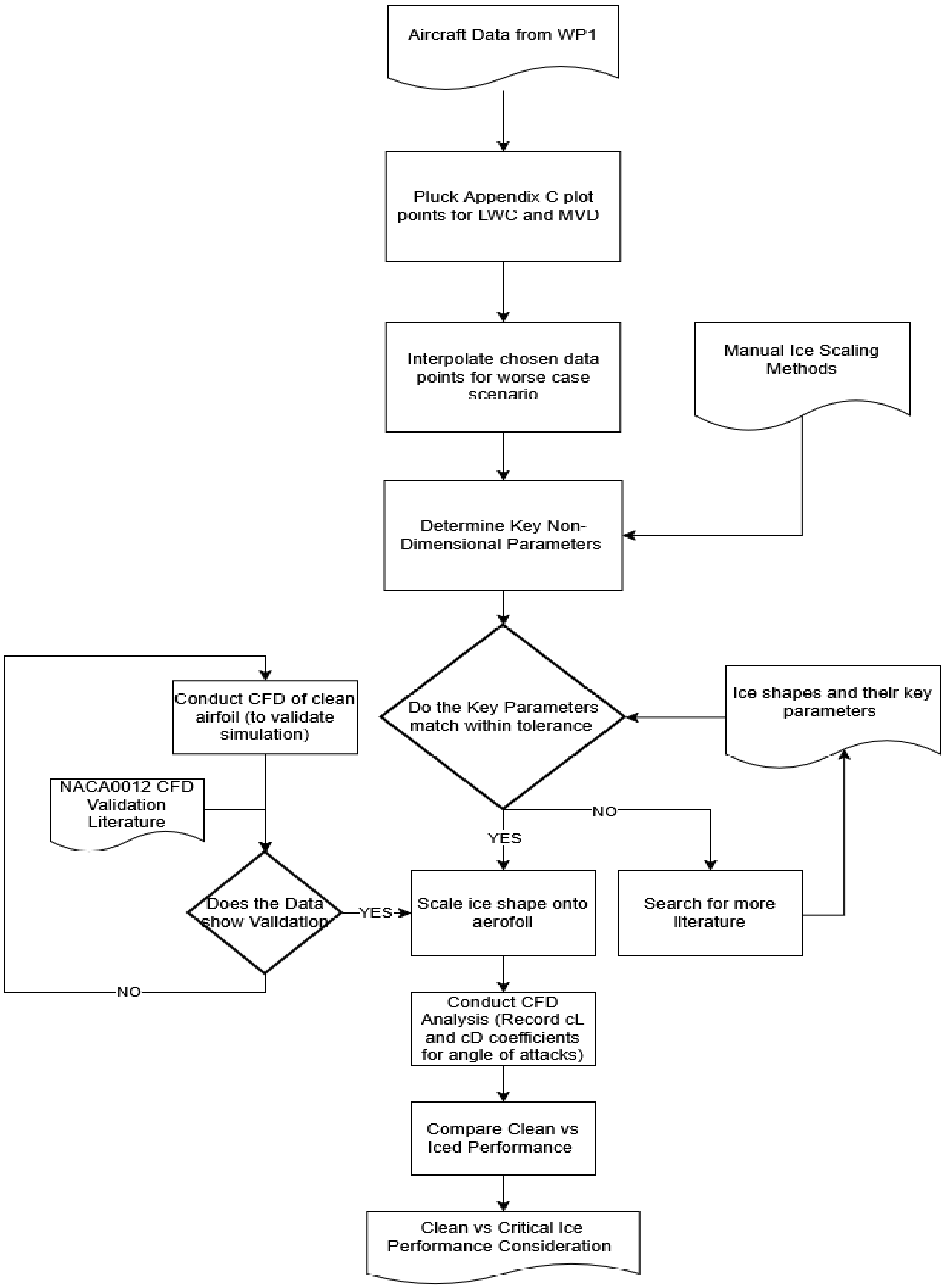

Figure 3: Case Study Methodology

The characteristic dimension, d, used throughout the scaling analysis was the cylinder diameter. The cylinder diameter is equal to twice the aerofoil leading edge radius. For a NACA0012 aerofoil, the leading edge radius equates to 0.0158c, where c is the chord length in metres [46]. Therefore, the characteristic dimension was calculated using:

$$d = 2 \times 0.0158c \tag{1}$$

Anderson recommends using characteristic dimension, d, rather than working with the full chord length, as the region near the leading edge of the aerofoil experiences the greatest ice accretion .

A static temperature of -8.5°C was chosen for the worst-case icing. Anderson clarified that the application of a total stagnation temperature greater than -3°C may lower the accuracy and reliability of the scaling method [45]. The stagnation temperature at holding airspeed, T_{total}, was calculated using the following equation:

$$T_{total} = T_{st} \times \frac{V^2}{2C_{p,a}} \tag{2}$$

Using **Equation 2**, the stagnation temperature at the holding airspeed was calculated to be -3.2°C, this value is below the recommended threshold given by Anderson while still remaining within the glaze ice regime, confirming the validity of the analysis.

A higher LWC results in a more critical ice formation, as higher amounts of energy are required to fracture the adhesive strength of the ice [28]. Although larger droplets possess higher collection efficiencies, smaller droplets containing higher LWC values exhibit a greater threat due to the promotion of glaze ice, leading to greater aerodynamic degradation.

LWC and MVD points for continuous and intermittent maximum conditions were extracted from the ASTM F3120/F3120M-20 standard data graphs. The chosen data points were interpolated to obtain an accurate LWC for the chosen static temperature. The obtained MVD and LWC points are shown below in **Table 2:**

Table 2: Liquid Water Content and Mean Volumetric Droplet Diameter selected from ASTM F3120/F3120M-20

Type of Icing Condition	MVD (μm)	LWC (g/m³)
Continuous Maximum (Stratiform Cloud)	15	0.63
Continuous Maximum (Stratiform Cloud)	25	0.33
Intermittent Maximum (Cumuliform Cloud)	15	2.56
Intermittent Maximum (Cumuliform Cloud)	20	2.19

There was no horizontal cloud extent correction factor, F, applied. This was because the ASTM F3120/F3120M-20 MOC document explicitly prohibits the use of the correction factor, F, when testing for this requirement [19].

The atmospheric and fluid properties observed at the analysis altitude of 10000ft were determined. The static pressure was obtained through the barometric formula for the troposphere:

$$P_{st} = P_0 \times \left(\frac{1 - L \times h}{T_0}\right)^{\frac{g}{R*L}} \tag{3}$$

The ideal gas law was used to determine the air density at the analysis altitude and temperature, ρ_{air}:

$$\rho_{air} = \frac{P_{st}}{R \times T_{st}} \tag{4}$$

The air density at the film temperature, ρ_{film}, was also determined:

$$\rho_{film} = \frac{P_{st}}{R \times T_{film}} \tag{5}$$

Using Sutherland's law, the air viscosity at the analysis altitude was determined:

$$\mu_{air} = \mu_{ref}\left(\frac{T_{st}}{T_{ref}}\right)^{1.5} \frac{T_{ref} + S}{T_{st} + S} \tag{6}$$

Subsequently, the air viscosity at the film temperature, μ_{film}, was found:

$$\mu_{film} = \mu_{ref}\left(\frac{T_{film}}{T_{ref}}\right)^{1.5} \frac{T_{ref} + S}{T + S} \tag{7}$$

The film temperature, defined as the mean temperature of the boundary layer, was applied to evaluate air viscosity, air viscosity at the film temperature, and air density at the film temperature. This ensured that the properties of the fluid near the surface were accounted for when determining the convective heat transfer.

The inertial parameter, *K,* was the first key similarity parameter determined. This parameter models the droplet trajectory, determining whether a droplet has sufficient momentum to strike the aircraft surface.

$$K = \frac{\rho_{water} \times MVD^2 \times V}{18 \times \mu_{air} \times d} \tag{8}$$

To account for actual aerodynamic drag, which is far greater than what the ideal Stokes' law predicts, the range parameter, *RP*, was determined:

$$RP = \frac{\lambda}{\lambda_{stokes}} = \frac{1}{0.8388 + 0.001483 \times (Re_{MVD}) + 0.1847 \times \sqrt{Re_{MVD}}} \tag{9}$$

The Reynolds number of the droplet is required to calculate the range parameter, *RP*, as illustrated through **Equation 9**. The expression for the Reynolds number is:

$$Re_{MVD} = \frac{\rho_{air} \times V \times MVD}{\mu_{air}} \tag{10}$$

The range parameter was combined with the inertial parameter, K, to produce a modified inertial parameter, K_0. This implements corrections for the ideal Stokes' law, increasing the accuracy of the scaling model. Anderson explained that the modified inertial parameter, K_0, when matched, guarantees accurate droplet trajectories over the entire aerofoil surface during wind tunnel testing [45].

$$K_0 = \frac{1}{8} + \frac{\lambda}{\lambda_{stokes}} \times \left(K - \frac{1}{8}\right) \tag{11}$$

The stagnation point collection efficiency, β_0, models the fraction of droplets that experience contact with the leading edge of the aerofoil. With OSCAR aerofoil featuring a sweep angle, Λ; the standard stagnation point collection efficiency formula was altered to the formula prescribed by Tsao [46]:

$$\beta_0 = \frac{1.40\left(K_0 - \frac{1}{8}\right)^{0.84}}{1 + 1.40\left(K_0 - \frac{1}{8}\right)^{0.84}}(cos\Lambda) \tag{12}$$

The accumulation parameter was the next key similarity parameter determined, this represents the maximum potential thickness of ice formed by the accumulation of droplets on the aerofoil [45]:

$$A_c = \frac{LWC \times V \times \tau}{\rho_{ice} \times d} \tag{13}$$

Where τ represents the exposure duration, which is 1350 seconds as defined by the ASTM F3120/F3120M-20 standard [19].

With the accumulation parameter defined, the last major similarity parameter to be determined is the freezing fraction. For glaze icing conditions, some impinging droplets are unable to freeze upon impact. To account for this, the energy balance at the surface of the aerofoil was first modelled.

The convective heat transfer coefficient at the stagnation point was determined. This involved calculating the thermal conductivity at the film temperature, Prandtl number, Reynolds number at the film, and the Nusselt number.

The thermal conductivity at the film temperature, k_{film}:

$$k_{film} = 1.7307 \times -0.008529 + 0.001016 * \sqrt{(T_{film} \times 1.8)} \tag{14}$$

The Prandtl number at the film temperature, Pr:

$$Pr = \frac{\mu_{film} \times C_{p,a}}{k_{film}} \tag{15}$$

Reynolds number at the film temperature, Re_{film}:

$$Re_{film} = \frac{\rho_{film} \times V \times d}{\mu_{film}} \tag{16}$$

Lastly, the Nusselt number at stagnation point:

$$Nu = 1.14 \times Pr^{0.4} \times Re_{film}^{0.5} \tag{17}$$

With the required properties defined, the convective heat transfer coefficient, h_c,was calculated. The swept angle modification created by Tsao was incorporated, as shown through **Equation 17**:

$$h_c = \frac{Nu \times k_{film}}{d} (cos\Lambda)^{\frac{1}{2}} \tag{18}$$

The gas phase mass transfer coefficient, h_G, needed to be defined. This coefficient is used to model the mass of water formed by the droplets that evaporate following impingement. This accounts for the total heat loss from evaporative cooling, maximising the accuracy of the obtained freezing fraction value.

The Schmidt number, *Sc*, at the film temperature was calculated first:

$$Sc = \frac{\mu_{a,film}}{\rho_{film} \times (2.11 * 10^{-5}) \times \left(\frac{T_{film}}{273.15}\right)^{1.94} \times \frac{101325}{P_{st}}} \tag{19}$$

The gas phase mass transfer coefficient was then determined:

$$h_G = \frac{h_c}{c_{p,a}} \left(\frac{Pr}{Sc}\right)^{0.67} \tag{20}$$

The relative heat factor, *b*, was assessed. This expresses the ratio between thermal capacity of the impinging droplets to the convective cooling ability of the aerofoil surface:

$$b = \frac{LWC \times V \times \beta_0 \times c_{p,ws}}{h_c} \tag{21}$$

Two energy transfer parameters quantify the energy transfer processes at the surface. The water energy transfer parameter which quantifies the thermal and kinetic energy of the impinging droplets. The air energy transfer parameter quantifies convection and evaporation principles [45],[43].

The formula used to calculate the water energy transfer parameter is shown below:

$$\phi = \left(T_{ref} - T_{st}\right) - \frac{V^2}{2c_{p,ws}} \tag{22}$$

To calculate the air energy transfer parameter, the vapour pressure at the surface temperature and the vapour pressure at the static temperature had to be computed first:

The vapour pressure at the surface temperature, p_{w_surf}:

$$p_{w\_surf} = 610.78 + 44.365\Delta T_s + 1.4289\Delta T_s^2 + 0.026506\Delta T_s^3 + 0.00030312\Delta T_s^4 + 0.0000020341\Delta T_s^5 + 0.0000000061368\Delta T_s^6 \tag{23}$$

Where: $\Delta T_s = T_{ref} - 273.15$

The vapour pressure at the static temperature, p_{w_st}:

$$p_{w\_st} = 610.78 + 44.365\Delta T_{st} + 1.4289\Delta T_{st}^2 + 0.026506\Delta T_{st}^3 + 0.00030312\Delta T_{st}^4 + 0.0000020341\Delta T_{st}^5 + 0.0000000061368\Delta T_{st}^6 \quad (24)$$

Where: $\Delta T_s = T_{st} - 273.15$

These two vapour pressures dictate the rate of evaporation of the runback water. Combining the resulting evaporative heat loss with the convective heat loss result in the complete air energy transfer parameter, which was calculated using **Equation 24** [43]**:**

$$\theta = \left(T_{ref} - T_{st}\right) - \frac{rV^2}{2c_{p,a}} + \frac{h_G}{h_c} \times \frac{\Lambda_v\left(p_{w_surf} - p_{w_st}\right)}{p_{st}} \quad (25)$$

Where r is the recovery factor. The recovery factor is a dimensionless parameter used in calculating the temperature of the air within the boundary layer with the recovered kinetic energy taken into account. At the stagnation point, the recovery factor equates to 1 [45].

The freezing fraction, n_0, is then obtained from the Messinger energy balance [45]:

$$n_0 = \frac{c_{p,ws}}{L_f}\left(\phi + \frac{\theta}{b}\right) \quad (26)$$

Where L_f is the latent heat of freezing. A freezing fraction below 1 describes the formation of glaze icing. A low freezing fraction value below unity corresponds to glaze, with lower values producing greater protruding horns and therefore more severe icing. A freezing fraction of 1 describes rime icing conditions [45].

As a result of the inclusion of a swept wing, with a sweep angle of 18.33°, modifications to the convective heat transfer coefficient and collection efficiency formulas were applied, in accordance with the methodology described by Tsao [46].

The droplet trajectory calculations utilised the freestream velocity. As the minimum holding speed is lower than Mach 0.3, the approach was deemed valid with the flow remaining within the incompressible regime [46].

The manual published by Anderson provides a means to determine the thickness of the accumulated ice at the leading edge, t_i. Although, the obtained ice thickness does not influence the similarity parameters used for scaling, it serves as an additional parameter used in determining the worst-case icing conditions. The formula to obtain the leading edge ice thickness, in millimetres, is shown below [45]:

$$t_i = 1000\beta_0 A_c n_0 d \quad (27)$$

The non dimensional parameters were calculated for each condition, with 2 testing points for each Appendix C condition. The freezing fraction and accumulation parameter were considered the biggest contributors to the selection of the worst-case condition. This is because these parameters possess the narrowest tolerance margins for scaling [45]. A lower freezing fraction and a higher accumulation parameter provided a more severe glaze ice formation, thereby increasing the overall criticality of the icing condition.

An ice shape was selected from existing literature for the worst-case continuous maximum condition. The ice shape is illustrated through **Figure 4** as “Run 414”. This ice shape was tested and documented within the scaling manual made by Anderson [45].

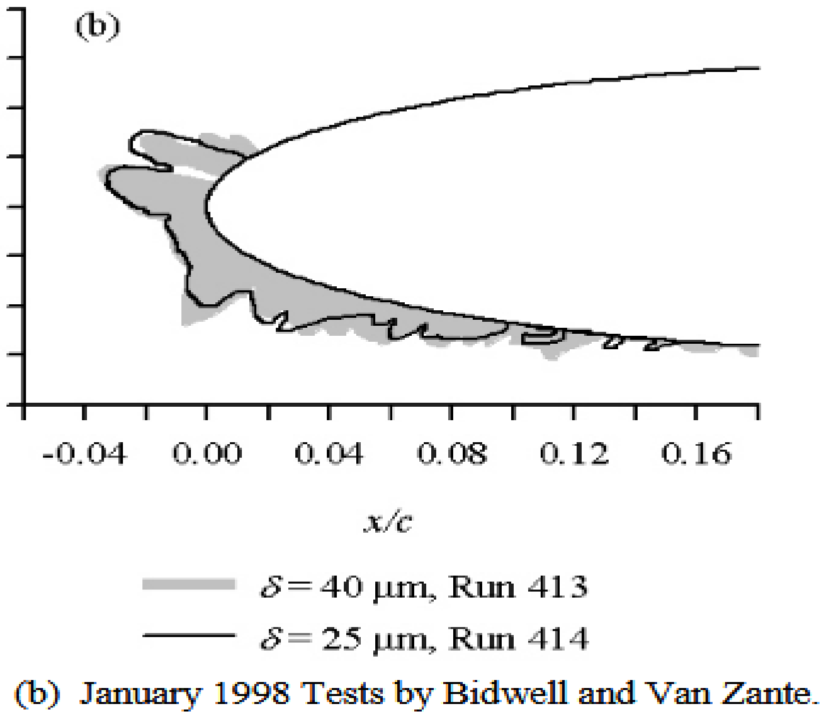

Figure 4: Ice Shape selected for Continuous Maximum Conditions [45]

The freezing fraction matched, with there being only a 1.25% difference. The accumulation parameter was also observed to be within the advisory 15% limit, showcasing a 13.7% difference. However, the obtained collection efficiency was substantially lower than the collection efficiency of Run 414, exhibiting a difference of 56.25%. No alternative ice shape in the available literature provided a closer match across all three parameters simultaneously. Therefore, the most influential parameters, judged to be the freezing fraction and the accumulation parameter were prioritised. This limitation was acknowledged by the project supervisor, who confirmed that the methodology undertaken and the choice made were the most appropriate option, and that analysis should be conducted with the chosen ice shape.

There was no available existing literature that provided an ice shape suitable for the worst-case intermittent condition, as the obtained freezing fraction yielded a value 50% lower than any available literature ice shape. Due to the lack of documented ice morphologies, suitable ice shapes for the worst-case intermittent icing parameters could not be identified. Thereby, preventing an accurate and reliable computational analysis. The worst-case intermittent maximum results were reported along with the calculated scaling parameters. **Table 3** displays the associated dimensional scaling parameters for the obtained ice shape.

The runback zone was determined by subtracting the calculated leading edge ice thickness for glaze conditions with the hypothetical leading edge ice thickness produced in rime icing conditions [45].

Table 3: Key scaling parameters for chosen ice shape [45]

Parameter	Run	*Chord, in*	T_{st}, *°F*	*V, mph*	*MVD, μm*	*LWC, g/m³*	τ, min	β_0, %	A_c	n_0
Value	414	21	13	230	25	0.55	7	80	1.54	0.48

3.3 Computational Fluid Dynamics Analysis

2D CFD analysis was performed in ANSYS Fluent to determine the aerodynamic degradation caused by the obtained ice shape for the worst-case continuous maximum icing conditions. The conducted analysis involved comparing the performance characteristics of a clean NACA 0012 aerofoil against an iced NACA 0012 aerofoil across a range of angles of attack. This section explores the process of modelling the ice shape onto the aerofoil, the mesh strategy, the simulation set-up, and the validation approach.

The clean NACA 0012 aerofoil geometry was generated using the Airfoil Tools online database. The geometric parameters of the aerofoil were inserted into the aerofoil plotter tool, based on the characteristics and dimensions displayed in **Table 1**, which produced a set of surface coordinates. The surface coordinates were exported as a CSV file, which was then reformatted into the ANSYS point curve format. The reformatted CSV file was imported into ANSYS Design Modeler within a Fluent system.

The iced NACA 0012 aerofoil was constructed in SOLIDWORKS. The CSV file was converted to a SOLIDWORKS curve file and imported. The selected ice shape obtained from the scaling manual was sketched onto the 2D aerofoil using the sketch picture tool. The ice shape was traced onto the leading edge of the aerofoil as shown by **Figure 5**:

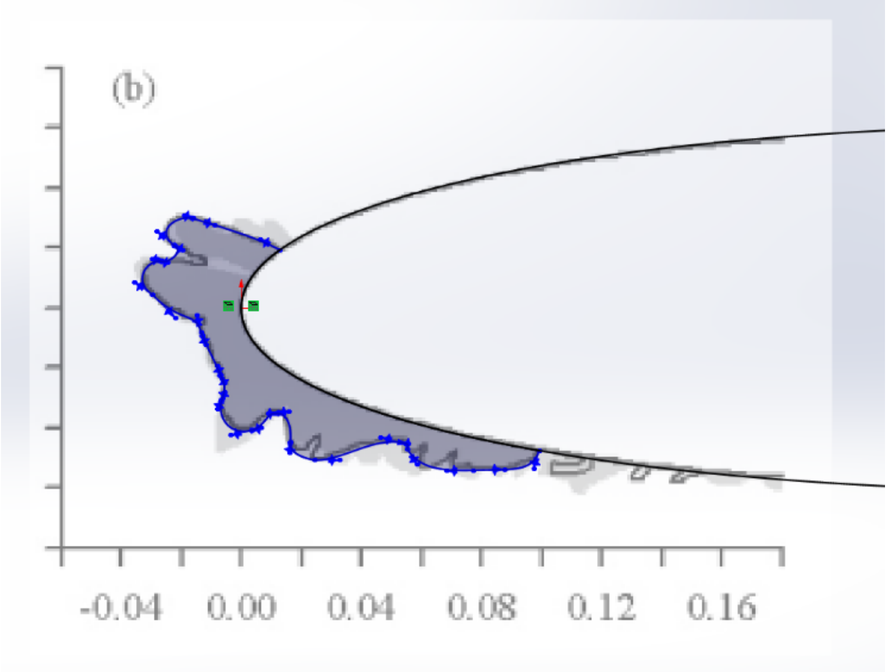

Figure 5: Traced out smooth ice shape in SOLIDWORKS

With the upper and lower horns of the chosen ice shape possessing irregular and sharp protrusions, meshing errors were encountered when replicating the characteristics of the horns. Therefore, to eliminate any errors that would be encountered during meshing, the protruding horns of the ice shape were traced smoothly, minimising the sharpness.

This adjustment in the tracing methodology was discussed with the project supervisor, who agreed with the proposition and advised that this can be considered as a conservative underestimation.

The feather ice elements present aft of the main ice accretion shape were not modelled. The primary objective of the CFD analysis was to obtain the aerodynamic penalties induced by the worst-case icing conditions. The aerodynamic degradation experienced by the aircraft is determined predominantly by the protrusion of the glaze ice horns. The incorporation of the feather ice elements would have added meshing complexity while offering minimal influence. This was acknowledged as another conservative estimate.

A C-type domain was used to define the computational domain rather than an O-type domain, as C-type domains capture gradients near the wake region with higher accuracy, making them suited for aerofoil simulation [47]. The domain extended approximately 9 chord lengths in all directions from the aerofoil.

For the clean aerofoil, the C-grid mesh comprised quadrilateral elements. This structure ensured accurate boundary layer resolution. **Figure 6** illustrates the structure of the mesh.

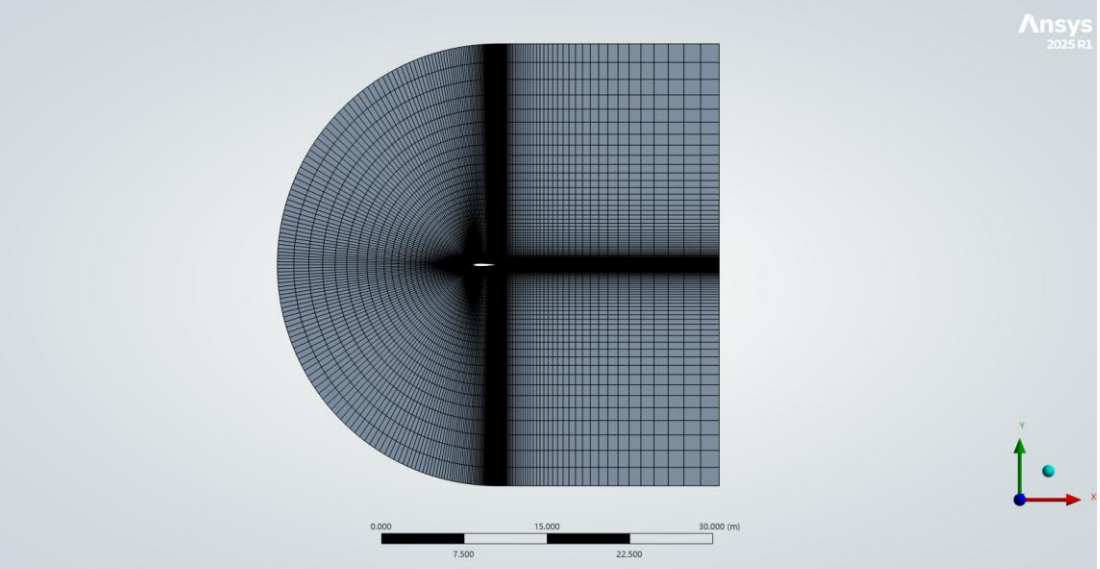

Figure 6: Mesh structure of the clean aerofoil

For the iced aerofoil, a fully unstructured mesh was employed. A fully structured mesh could not be implemented; the complex ice shape caused excessive meshing errors with a high y+ value. An attempt was made to split the domain into two bodies, with the inner body being unstructured and the outer body being structured. However, due to constraints in ANSYS Meshing, this change was not possible.

The finalised iced aerofoil mesh comprised approximately 101,000 nodes and 125,000 elements. The mesh was refined using the inflation and edge sizing tool. Edge sizing was implemented on the clean aerofoil region, and the inflation tool was used to refine the ice shape region. Smooth Transition was used instead of First Layer Thickness, as the First Layer Thickness produced layer collisions in the concavities between the ice features. Unfortunately, a mesh convergence study could not be conducted on the iced aerofoil as a result of time constraints.

Table 4 summarises key mesh quality metrics, with all values within acceptable ranges for unstructured meshes modelling complex geometries.

Table 4: Key Mesh Quality Metrics

Metrics	Value
Minimum Orthogonal Quality	0.219
Maximum Aspect Ratio	221
Average y+ (Area Weighted)	1.9
Y+ on clean surface	~1
Y+ on iced shape	~2.5-3

The clean and iced aerofoil simulations were both set up using the Spalart-Allmaras (SA) turbulence model; this ensured consistency between comparisons. Spalart-Allmaras is a one equation model specialised for aerofoil simulation and boundary layer analysis that solves for the modified turbulent viscosity using the transport equation. This model consumes less computational power than other aerofoil specialised models such as k-omega SST, while also remaining numerically stable when simulating unstructured meshes. The mesh structure of the clean aerofoil was validated against published case data [48].

Given the high Reynolds number for this case study, the validation was conducted on different boundary conditions to ensure the mesh was capturing accurate data.

A steady state pressure-based solver was employed for both clean and iced aerofoil simulations. The analysis air density and viscosity were entered into the fluid properties menu, with both categories set to constant. A coupled pressure-velocity scheme was utilised. For the spatial discretisation: pressure was set to second order, and momentum and modified turbulent viscosity were set to second order upwind. The freestream boundary conditions were set to match the holding conditions at the analysis altitude. Turbulence viscosity ratio was set as the specification method at the inlet, with the turbulent viscosity ratio set to 1. The x and y components of the flow direction were modified in accordance with the angle of attack. Each simulation was performed with 5000 iterations per angle of attack case, with angles of attack ranging from 0° to 17°. Convergence was assessed with the residuals demanding an absolute criteria of 10^{-6}.

The wall roughness model was set to standard for both configurations, with there being no roughness applied. The high roughness (icing) model was not selected for the iced aerofoil. This decision stemmed from the mismatch between the calculated collection efficiency and the literature collection efficiency. With the collection efficiency being low, the roughness constant provided throughout the existing available literature was unable to be matched for the obtained collection efficiency value. Randomly allocating a roughness constant based on limited known parameters and mismatched collection efficiency values would only increase the likelihood of producing inaccurate results. Additionally, the roughness model used in ANSYS Fluent requires the first mesh cell to be situated above the roughness elements. This is incompatible with the y+ value obtained by the mesh used in the analysis, thereby validating the methodology carried out for the setup.

Figures 7 and **8** show the unstructured triangular mesh imported into Fluent, showcasing the element distribution around the aerofoil and a close up view of the iced region respectively. The successful translation of the mesh into Fluent further validates the accuracy of the solution.

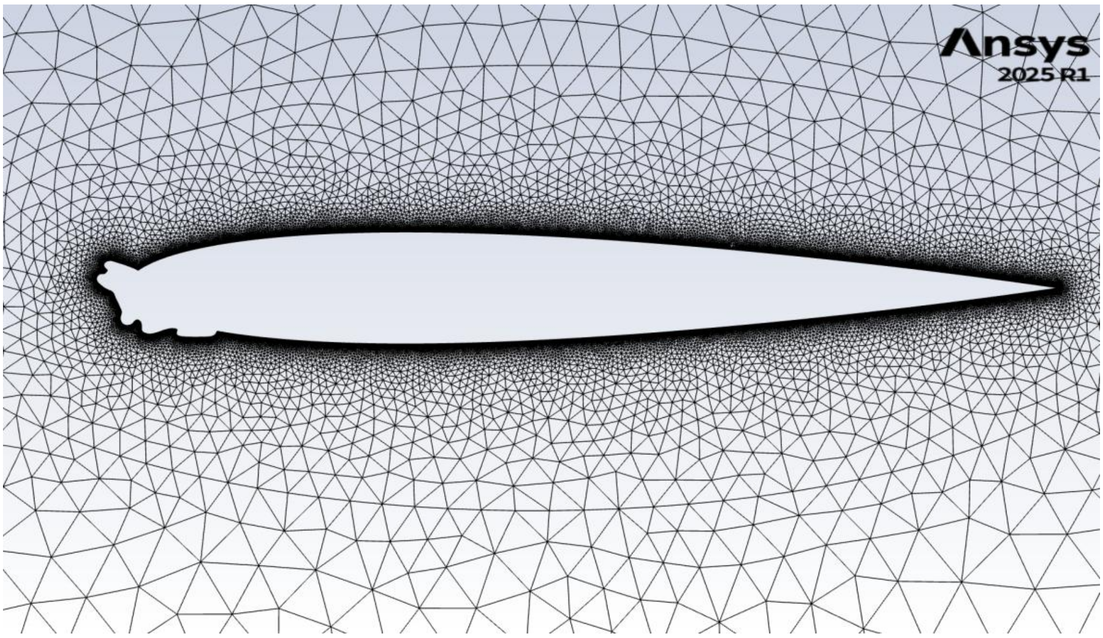

Figure 7: Computational unstructured triangular mesh around the iced NACA 0012 aerofoil, displayed in ANSYS Fluent.

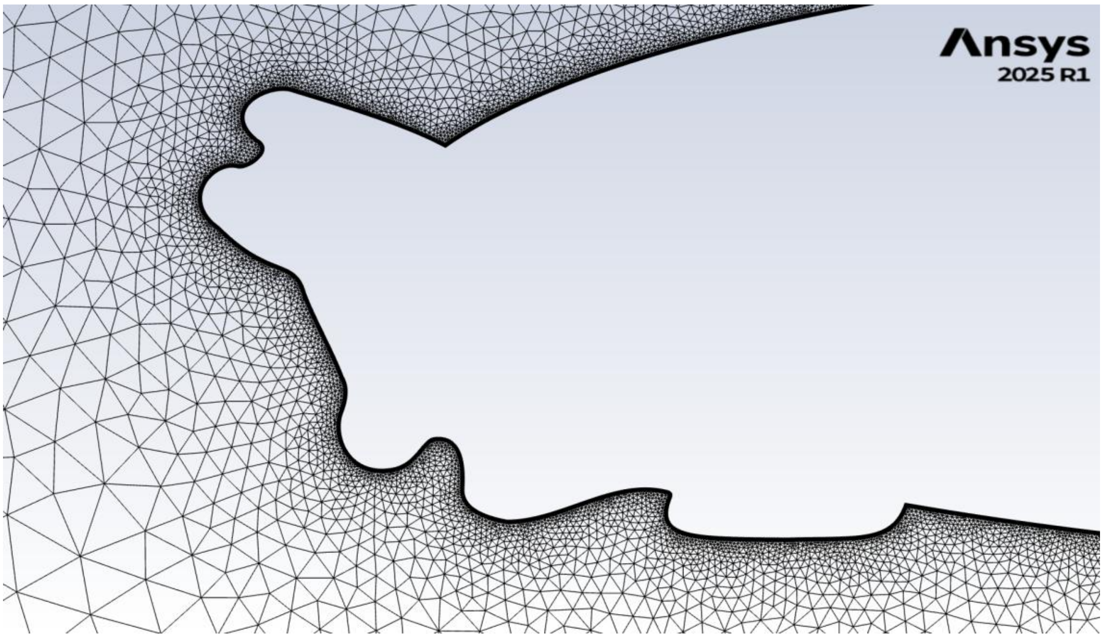

Figure 8:Close-up view of the ice region at the leading edge of the aerofoil, showing mesh refinement around the ice horns and concavities.

3.4 Anti-Icing and De-Icing Technology Comparative Assessment

An analysis of the various technologies used to combat in-flight icing was performed. The technologies examined include active and passive ice protection systems, and ice detection systems. The objective of the analysis was to provide the OSCAR design team with a list of several technologies, highlighting their strengths and weaknesses, as well as assessing the impact on certification compliance when adopting each technology.

The evaluation of the technologies included the following criteria:

Energy Consumption: SC-VTOL.2430 mandates that the energy storage system be capable of providing sufficient reserves, even under icing loads. This criterion was weighted heavily due to its regulatory significance.

Weight Penalty: The additional mass added by the chosen technology. This was evaluated in the context of the aircraft MTOW and payload requirements

CFRP Compatibility: With the design team opting for an aircraft made of CFRP, it was important to evaluate the effectiveness of each technology when integrated with or applied to the aircraft. The effects on structural integrity were also considered.

Certification Alignment: The maturity of the technology relative to ASTM F3120 and the broader certification framework. Technologies that have a proven record in achieving icing certification were favoured. The influence of each technology in satisfying specific requirements was also considered.

Erosion Resistance: The durability of each technology, specifically against rain erosion, was determined. This criterion was emphasised when conducting analysis on passive coatings.

Manufacturability: Evaluating the manufacturing costs and complexities of each technology, prioritising low cost solutions.

Each technology was assessed qualitatively against these six criteria using data provided by the literature. Quantitative performance data relating to the criteria was recorded if available. Furthermore, an analysis on commercially available technologies was also conducted, which aimed to satisfy the requirements outlined by the Airbus supervisors.

The data gathered from each technology was sorted into a matrix to facilitate comparisons between different technologies. As the design of the OSCAR aircraft matures, the creation of various aircraft configurations will lead to variances in the behaviour and characteristics in specific regions. While design decisions solely rest with the design team, WP4 is positioned to provide recommendations informed by the certification analysis. The methodology for the ice technology comparative assessment is illustrated in **Figure 9:**

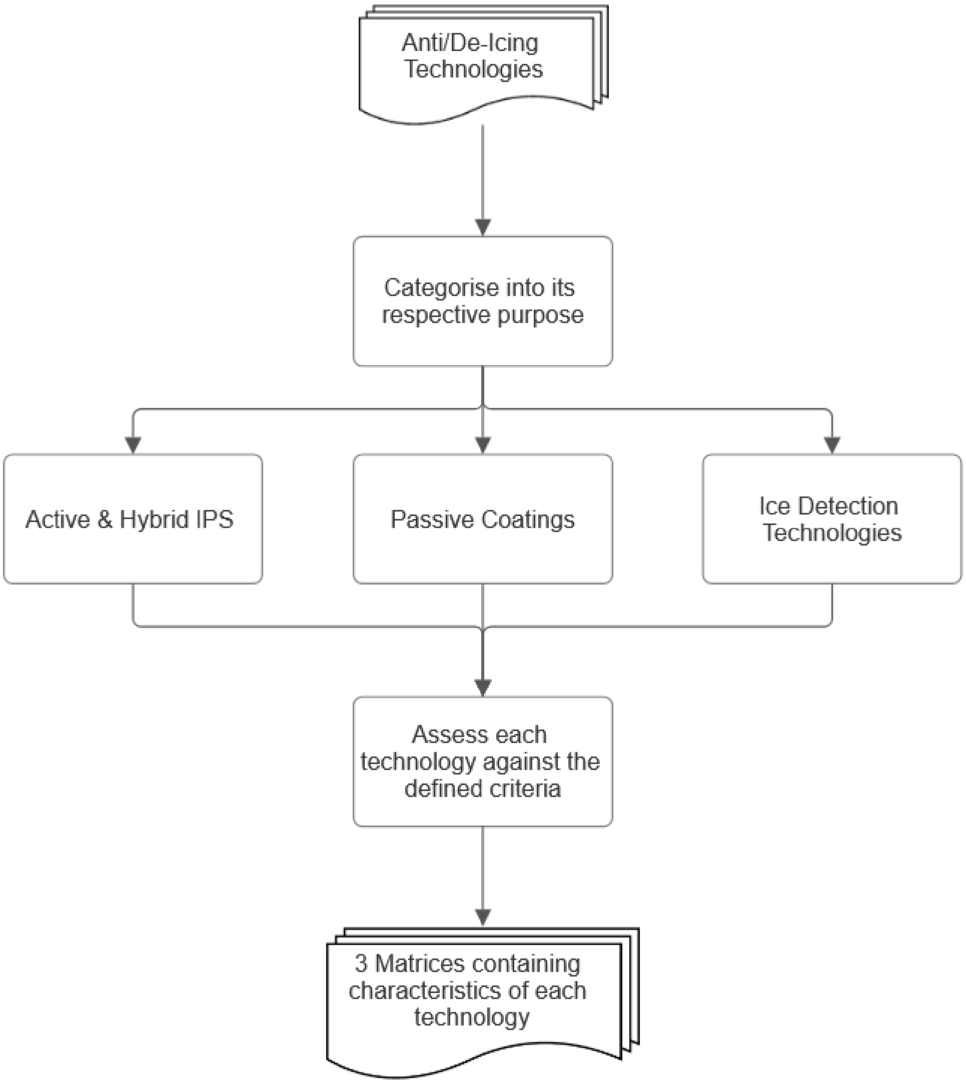

Figure 9: Methodology undertaken for the ice technology comparative Assessment and the construction of the matrices

4.0 Results and Discussion

This section presents and discusses the results of the certification compliance matrix, the ice scaling case study and its associated CFD analysis, the anti-icing and de-icing technology comparative assessment, and the traceability of each deliverable across the OSCAR programme.

4.1 Certification Compliance Matrix

The constructed certification matrix consolidates all the Appendix C icing related requirements across four regulatory frameworks, establishing the regulatory envelope against which the other project deliverables were analysed. The certification compliance matrix was organised in two layers. The upper layer maps 13 distinct certification specification (CS) and special condition (SC) requirements, with each requirement linked to its associated means of compliance (MOC) standard. The lower layer expands each MOC compliance standard, dividing and organising the content into individual sub-requirements, each paired with various testing methodologies. A single upper layer requirement may prescribe multiple MOC documents rather than one, yielding 17 requirement-to-MOC mappings. The matrix preserves each mapping rather than collapsing them, ensuring that any revisions to the certification specification can be implemented into the matrix with minimal difficulties.

The lower layer boasts 110 sub-requirements mandated by the five external MOC standard specifications. The distribution of requirements across the applicable MOC standards is expressed in **Table 5:**

Table 5: Distribution of icing sub-requirements across relevant MOC standards

MOC Specification Standard	Number of Associated Sub-Requirements
ASTM F3120/F3120M-20	62
ASTM F3316/F3316M-19	9
ASTM F3062/F3062M-20	2
ASTM F3066/F3066M-18	13
EUROCAE ED-314	23

CS-23 (Amendment 6) contains three top-level requirements: CS 23.2165 (performance and flight characteristics in icing), CS 23.2415 (powerplant installation ice protection) and CS 23.2540 (flight into known icing).

SC-VTOL (Issue 2) contains two top-level requirements: VTOL.2165 (flight in icing conditions) and VTOL.2415 (lift/thrust system installation ice protection). Although both VTOL.2165 and CS 23.2540 may seem identical, VTOL.2165 considers solely inadvertent icing encounters.

SC E-19 EHPS (Issue 1) contains five top-level requirements: EHPS.80 (safety assessment), EHPS.280 (icing and snow conditions), EHPS.300 (fuel system), EHPS.350 (electric-hybrid propulsion control system) and EHPS.460 (operational demonstration). SC E-19 EHPS referenced no specific means of compliance standard or methodology.

It is implicitly required that the aircraft demonstrates safe operation throughout the specified icing conditions for which certification is sought, ensuring all the components mentioned in the requirements remain safe from failure. The five top-level requirements were linked to a MOC specification standard which encompassed the CS/SC requirement.

CS-P (Amendment 2) contains three top-level requirements: CS-P 350c (Centrifugal Load Test), CS-P 380 (Lightning Strike) and CS-P 390 (Endurance Testing). Unlike the previous three regulatory frameworks, CS-P contained the relevant MOC sub-requirement(s) within the certification specification document, rather than referencing an external specification made by a third party such as ASTM or EUROCAE. With these MOC requirements being internal to the CS-P framework, there was no influence on the sub-requirements tally, explaining its absence from **Table 5.**

The justification for characterising the matrix into two distinct layers is to preserve and strengthen the modularity of the links between the certification requirement and means of compliance documents. This approach was demonstrated clearly for the requirement CS 23.2415, which requests "The airplane design and powerplant installation design must prevent predicted ice accretion and ice shedding that would harm the powerplant operation within the icing conditions for which certification is requested". This requirement maps to multiple means of compliance standards. F3120/F3120M-20 addresses the ice protection functionality and ice shedding requirements. F3062/F3062M-20 addresses requirements solely applicable to carburetor de-icing fluid systems. F3066/F3066M-18 addresses requirements directed at turbine engine ice protection and induction system ice protection. Lastly, F3063/F3063M-20 provides requirements to which the fuel system must adhere in order for F3062/F3062M-20 testing to be conducted. Similarly, EHPS.280 maps to both F3120/F3120M-20 and EUROCAE ED-314.

A design team reading these clauses in isolation would be able to easily visualise what the certification specification requires and what individual requirements need to be fulfilled and tested for. Furthermore, the separation of each MOC document into individual worksheets instead of compiling all the requirements into a single table further simplified the matrix when facilitating the hand off: incoming OSCAR candidates can extend or revise the compliance requirements within an individual MOC document without altering the structure of the other documents, and any future revision to a MOC document, such as ASTM or EUROCAE, can be updated through the alteration of a single worksheet rather than editing a monolithic table.

The sub-requirements listed within each MOC document were connected with one or more applicable testing methodologies, which were selected from the testing provisions supplied within the MOC standards themselves. F3120/F3120M-20 provided the greatest contribution to the testing methodologies adopted, highlighting certification proven testing methodologies such as ice wind tunnel testing and computational fluid dynamics analysis.

As shown in **Table 5**, ASTM F3120/F3120M-20 supplied the greatest number of sub-requirements, with 62 sub-requirements developed. For simplicity, the MOC matrix for ASTM F3120/F3120M-20 comprised various sections. The main section comprised the requirements for aircraft components, such as the windshield, engine inlet, pitot and static probes, IPS and ice detection system.

The Annex A1 section covers flight test requirements in specified icing conditions which the aircraft must adhere to; these include climb after partial loss of thrust, en route climb and descent, approach and baulked landing, ICTS resistance, lateral control reversibility using bank rolls, and autopilot operation (if equipped).

The Annex A4 section lists specific ice detection requirements, consisting of primary and advisory system classification, nuisance activation prevention, temperature threshold accuracy and manual activation delay testing.

EUROCAE ED-314 primarily encompassed VTOL specific requirements that CS-23 did not specify. The most notable requirements were the five minute inadvertent icing exit requirement and the AFM content specification. The latter requirement was used in the creation of a sample AFM which includes flight procedures necessary for the pilot training program. The sample AFM was fed to the services team (WP3), who are responsible for designing both the pilot training program and the flight simulator.

Ice wind tunnel (IWT) testing and computational fluid dynamics analysis, with ANSYS FENSAP-ICE and NASA LEWICE being the most cited codes in the F3120 testing provisions, dominated the testing methodologies within the multiple MOC matrices. Flight testing is applicable for all requirements; however, due to high costs, this testing method was only assigned for large scale components requiring complex testing conditions and aircraft flight characteristic specific requirements. As mentioned in **Section 3.1**, multiple testing methodologies facilitate cost and time effective testing strategies, where several sub-requirements can be evaluated within a single testing campaign. For instance, a dry-air flight test using the literature iced shape geometry shown in **Section 3.2** can simultaneously satisfy the climb requirements listed in the Annex A1 section, ICTS demonstration, stall warning verification and lateral control requirements. These testing methods form the starting point from which a consolidated testing plan can be built, with the later additions of time and costs estimates as the project matures.

It is important to note that there were several limitations encountered during the construction of the matrix which should be acknowledged. EUROCAE ED-103A which outlines the minimum operational performance specifications (MOPS) for active ice detection systems used during in-flight operations, could not be accessed. Although this standard is not mandated by EASA for CS-23, its performance criteria would have been implemented in all the certification specification requirements involving an ice detection system, refining the ice detection requirements against CS 23.2165, CS 23.2415 and VTOL.2165. Additionally, the design team would be able to view the performance requirements in the form of a mandatory benchmark to base the design of the aircraft on. The SAE AS5562 (Ice and Rain Minimum Qualification Standards for Pitot and Pitot-Static Probes) standard could not be accessed on the same basis. The addition of SAE AS5562 would have refined the pitot probe specific sub-requirements listed in the MOC matrices through the addition of numerical thresholds. The University of Strathclyde library does not possess any copies of these standards; procurement and receiving approval for the standards was not possible due to time constraints. Acquiring ED-103A and SAE AS5562 should be prioritised by the future OSCAR candidates to further refine the matrix.

The top-level certification matrix is displayed in **Appendix A**. The sub-requirements could not be displayed due to copyright restrictions.

4.2 Case Study: Ice Shape Scaling and Aerodynamic Degradation

The worst-case ice shape parameters were obtained through the adoption of the methodology described in **Section 3.2**. Four Appendix C conditions were sampled at a static temperature of -8.5°C and an analysis altitude of 10000 ft, consisting of two stratiform cases and two cumuliform cases. The analysis concentrated on collecting data points with high LWCs, which resulted in relatively small mean volumetric diameters (MVD). The LWC for each condition was obtained through the Appendix C envelope graphs, documented in the ASTM F3120/F3120M-20 standard, and interpolated for values at the desired static temperature. The obtained and tested LWC and MVD points are shown in **Table 6:**

Table 6: Scaling Parameter Outputs for the tested Appendix C Conditions

Condition	*MVD, μm*	*LWC, g/m³*	K_0	β_0	A_c	n_0	t_i, mm	*Runback, mm*
Continuous Max 1 (CM1)	15	0.63	0.478	0.350	1.326	0.487	16.25	17.14
Continuous Max 2 (CM2)	25	0.335	0.999	0.556	0.705	0.560	15.77	12.4
Intermittent Max 1 (IM1)	15	2.559	0.478	0.350	5.385	0.189	25.59	109.98
Intermittent Max 2 (IM2)	20	2.198	0.721	0.451	4.627	0.179	26.94	123.32

The selection of the worst-case icing scenario was primarily determined by the freezing fraction and accumulation parameter, as these two similarity parameters have the narrowest tolerance margins, at 10% and 15% respectively. For glaze ice morphology, a lower freezing fraction coupled with a high accumulation parameter represents the most critical icing scenario. The thickness of the ice formed at the leading edge of the aerofoil influences the magnitude of the aerodynamic degradation, with a greater ice thickness yielding a more severe aerodynamic penalty.

Using the defined criteria, CM1 was judged as the most severe icing scenario for continuous maximum conditions. CM1 boasts a substantially higher accumulation parameter when compared with CM2, with an obtained value of 1.326, which is 88.1% greater than the value calculated for CM2. Additionally, CM1 possesses a lower freezing fraction than CM2, though not as substantial as the difference between the accumulation parameters. Although CM2 has a greater collection efficiency, the collection efficiency was judged to be less influential than the freezing fraction or accumulation parameter when determining the worst-case icing scenario. This assumption was validated through the thickness of the ice formed at the leading edge, with CM1 producing a thicker ice shape, resulting in more severe aerodynamic drag.

Applying the same assumptions made for the continuous maximum condition assessment for the intermittent maximum conditions, IM2 was assessed to produce the most severe ice scenario. Although IM1 possesses a significantly greater accumulation parameter, IM2 produces a lower freezing fraction and a thicker ice shape at the leading edge.

A lower freezing fraction coupled with a thicker ice shape was deemed to induce a greater penalty than a higher accumulation parameter due to the increased flow separation experienced in the IM2 scenario, resulting in a higher stall velocity.

When comparing a severity analysis between CM1 and IM2, IM2 exhibits more severe icing as illustrated by the values shown in **Table 6**. For the IM2 scenario, a freezing fraction of 0.179 indicates that less than a fifth of the impinging droplets froze upon impact, with the remaining droplets contributing to the formation of runback icing. The runback zone spans 123.3 mm aft of the stagnation point for the IM2 condition, promoting flow separation and making the intermittent icing condition more hazardous than the continuous condition.

The calculated continuous worst-case icing scenario compared with Anderson's Run 414 ice shape. **Table 7** shows the comparison between the calculated continuous maximum parameters (CM1) and the literature ice shape:

Table 7: Comparison between calculated continuous and literature ice shape similarity parameters

Parameter	Calculated Continuous (CM1)	Literature Run 414	% Difference
β_0	0.35	0.8	56.25
A_c	1.326	1.54	13.90
n_0	0.487	0.48	1.46

As illustrated by **Table 7**, the freezing fraction displayed excellent similarity, with a low % difference which is within Anderson's 10% tolerance margin. This suggests that the proportions of the ice shape that are formed at the leading edge of the aerofoil in the literature case match the proportions of the ice shape for the calculated continuous scenario (CM1). Matching proportions for the ice shape imply matching horn shapes, ensuring the flow separation behaviour captured in the simulation is accurate to the behaviour for CM1.

The collection efficiency for CM1 is substantially lower than the collection efficiency for the literature case, exhibiting a 56.25% difference. This mismatch was attributed to the exposure times differing significantly for both conditions rather than a calculation error. CM1 assumes an exposure time of 22.5 minutes, as mandated by ASTM F3120/F3120M-20 Annex A2.3 [19]. Run 414 has an exposure time of 7 minutes which is significantly lower than CM1. The difference in MAC size may have also contributed to the mismatched collection efficiency, with a larger chord typically yielding a lower collection efficiency in part due to decreased streamline curvature. A lower calculated collection efficiency translates to decreased water impingement per unit surface area, suggesting an inflated ice shape size for the literature ice shape. This was treated as conservatism that favours the safety case, with a bigger ice shape producing a larger aerodynamic penalty.

There was no available literature ice shape for the worst-case intermittent icing condition (IM2). There was a roughly 50% difference between the calculated freezing fraction and the lowest freezing fraction recorded for any NACA 0012 ice shape in the NASA Glenn database or any other available literature.

This margin was deemed too significant to yield an accurate representation of the aerodynamic degradation experienced for IM2 conditions; this implication is significant and should be flagged as high priority for the future OSCAR candidates. Access to specialised CFD icing software such as ANSYS FENSAP-ICE should be requested in order to obtain reliable and accurate flight penalties induced by this icing condition.

Comprehensive calculations for all the tested Appendix C icing conditions are provided in **Appendix B.**

4.3 Computational Fluid Dynamics Analysis

The aerodynamic degradation of the ice shape obtained from Run 414 was sketched onto an aerofoil, and the resulting flight characteristics were investigated through 2D CFD analysis using ANSYS Fluent. The Spalart-Allmaras (SA) model was chosen; the rationale is given in **Section 3.3.** Both aerofoil configurations were tested at a Reynolds number of 12.84×10^6 for angles of attack of 0° to 18° in increments of 2°. Additional single degree points were captured at an angle of attack of 7° and 17° to better characterise the stall region for both aerofoil configurations. The iced configuration was simulated under second order upwind discretisation schemes up to the stall angle, then under first order upwind with reduced under-relaxation factors from the stall angle onwards to maintain convergence and observe post-stall behaviour.

The results of the clean NACA 0012 aerofoil are displayed in **Table 8.** The clean configuration experienced steadily rising coefficients of drag (Cd) from angles of attack of 0° to 8°. From angles of attack of 8° to 18° the Cd underwent an exponential increase, specifically in the post-stall region where the Cd reaches a value of 0.212, shown by **Figure 10**. The coefficient of lift (Cl) experienced steady linear growth from angles of attack of 0° to 14°, with a sharp decline in Cl to 0.68 at an angle of attack of 18°. The clean configuration stalls at an angle of attack of 16° as illustrated in **Figure 11**, which is consistent with the expected behaviour of a thin symmetric aerofoil simulated under a high Reynolds number. The clean configuration serves as the baseline against which the iced aerofoil is compared.

The iced configuration behaved significantly differently from the clean configuration across the full angle of attack scope. **Table 8** illustrates the clean and iced lift and drag coefficients under second order upwind discretisation schemes. At an angle of attack of 0°, the iced aerofoil produced an adverse Cl of -0.002, whereas the clean configuration displayed a Cl of 0. The lower horn of the iced aerofoil protrudes further than the upper horn, establishing a negative camber effect, altering the behaviour of the pressure and inducing an adverse lift coefficient. At the cruise angle of attack of 4°, Cl experienced a 14% reduction, dropping from 0.437 (for the clean aerofoil) to 0.374 (for the iced aerofoil). Cd experienced a substantial increase from 0.0088 to 0.0237, which is a 169% increase. These values are aligned with typical glaze icing degradation estimates, which was explored in **Section 2.1**

Table 8: Clean and Iced Lift and Drag Coefficients

Angle of Attack (°)	Clean Cl	Iced Cl	Clean Cd	Iced Cd	ΔCl (%)	ΔCd (%)
0	0	-0.002	0.0074	0.014	0	89.2
2	0.22	0.1945	0.0085	0.0166	-11.4	95.3
4	0.437	0.374	0.0088	0.0237	-14.4	169.3
6	0.652	0.5253	0.0106	0.0378	-19.4	256.6
7	0.7583	0.5574	0.0118	0.0512	-26.5	333.9
8	0.86	Divergence	0.0132	Divergence	N/A	
10	1.062		0.0169			
12	1.2447		0.0222			
14	1.3945		0.0302			
15	1.4441		0.0365			
16	1.456		0.0465			
17	1.3436		0.0713			
18	0.68		0.212			

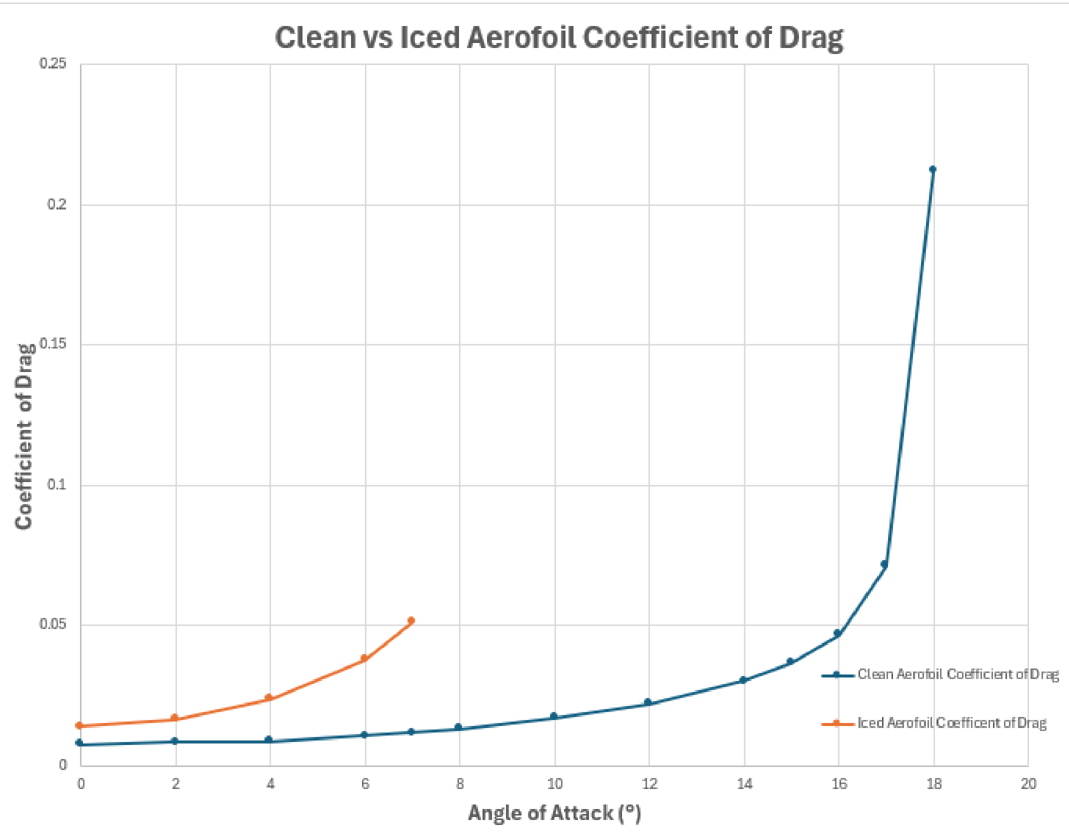

Figure 10: Clean vs Iced Aerofoil Coefficient of Drag mapped against Angle of Attack

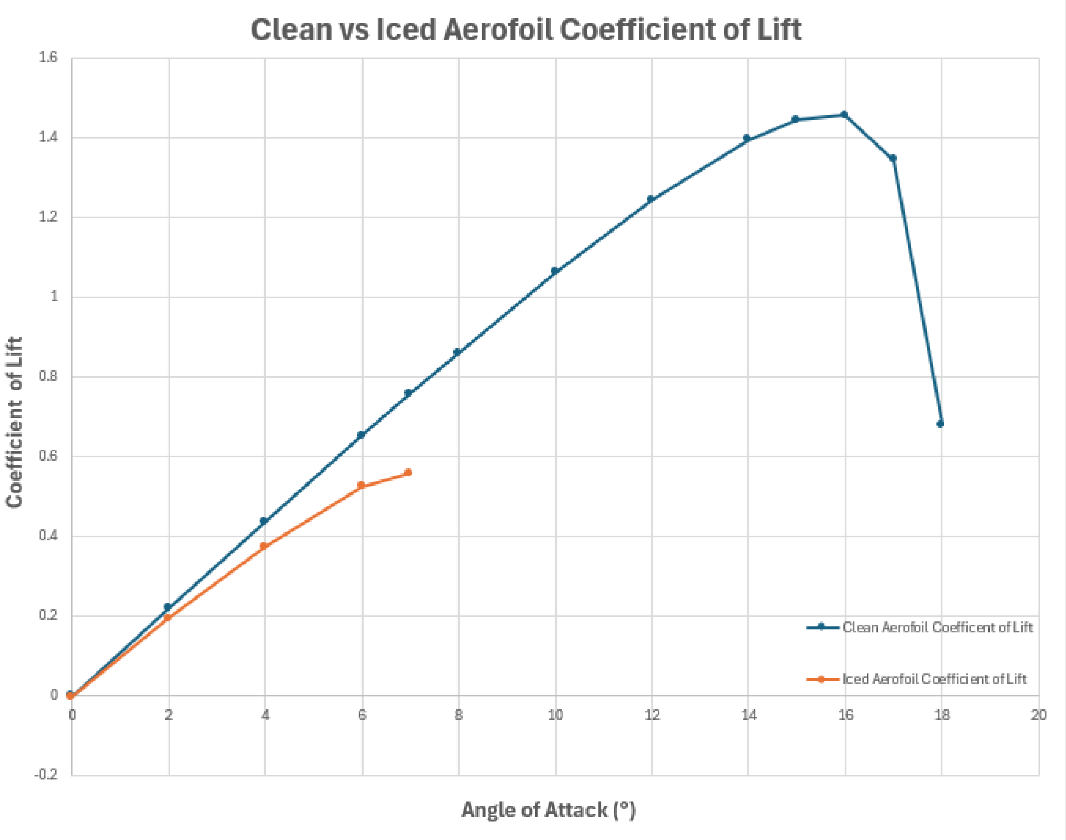

Figure 11: Clean vs Iced Aerofoil Coefficient of Lift mapped against Angle of Attack

The 14.4% loss in Cl and 169.3% increase in Cd at the cruise angle of attack (4°) are the main aerodynamic penalties to share with the design team as the OSCAR aircraft primarily operates at this angle of attack. The rate of degradation intensified with the percentage difference between the clean and iced aerofoil widening. This is observed for the Cl experiencing a 19.4% drop and a surge in Cd by 256.6% when comparing the iced configuration to the clean configuration for an angle of attack of 6°. The 7° angle of attack entry represents the last angle at which second order convergence could be achieved. The Cd experienced a massive surge, showing a 333.9% increase in the percentage difference. The Cl exhibited a 26.5% decrease, showing the beginning of the lift coefficient curve flattening which characterises the emergence of flow separation, occurring as a result of stall.

The disparity between the Cd and Cl percentage penalties at the cruising angle of attack (169.3% for Cd against 14.4% for Cl) demonstrates the effect the simulated ice shape has on pressure drag. The upper and lower horns of the ice shape behave as a small bluff body producing a stagnation region on its windward face and a pressure differential across its wake, which makes up the pressure drag mechanism found in the iced configuration.

Figure 12 displays the concentration of the pressure coefficient across the ice shape at the cruising angle of attack, illustrating the bluff body mechanism. A deep suction region is situated at the tip of the upper horn, with a minimum c_p of -3.85, opposed by a high pressure stagnation region at the forebody of the aerofoil, producing a c_p of approximately 1.

Whereas the lift penalty is comprised of the degradation of the suction side pressure distribution. At the cruise angle of attack, the lift to drag ratio falls from 49.7 for the clean aerofoil to 15.8 for the iced aerofoil, which equates to a 68.2% reduction in the recorded aerodynamic efficiency, illustrating a severe reduction in aircraft performance, drastically compromising fuel economy and manoeuvrability.

Figure 12: Pressure Coefficient Concentration across Ice Shape at cruise angle of attack

The steady state RANS solver was unable to achieve convergence for simulations of angles of attack beyond 7°, as a result of the intensity of the flow separation. The force monitors yielded rabidly oscillating results, unable to settle within the 5000 iteration budget. The iterations were increased to 20,000; however, this had no effect on the force monitor outputs. The discretisation scheme was lowered to first order upwind, which produced convergent results. However, the artificial numerical diffusion mechanism present in first order schemes suppressed the flow separation behaviour. This prevented an accurate stall angle of attack from identified; therefore, these results were excluded from **Table 8.** A transient (URANS) simulation was attempted but the computational demands imposed were beyond the available hardware capabilities. Therefore, the stall angle for the iced configuration was concluded to be at 7°, which represents a reduction of 9° from the clean stall angle of 16°.

The behaviour of the airflow across the iced aerofoil was visualised through pressure coefficient contours, velocity contours and turbulent viscosity ratio contours taken at the cruise and stall angles of attack (4° and 7° respectively).

At an angle of attack of 7°, the suction region at the tip of the upper horn intensifies to c_p of -3.97 and is concentrated further down the horn when compared to pressure coefficient displayed in **Figure 12**. An adverse pressure gradient is formed on the other upper horn, though not as concentrated as the upper horn, achieving a c_p of ~ -1.49, as shown in **Figure 13**. The steeper adverse pressure gradient induced by the glaze horns provokes boundary layer separation at lower angles of attack, resulting in a surge in drag and decline in lift.

Figure 13: Pressure Coefficient Concentration across Ice Shape at stall angle of attack

At the cruising angles of attack, the flow stagnates on the forepart of the glaze ice horns, accelerating sharply at the horn tip to a peak velocity of 204 m/s as shown in **Figure 14**. Immediately aft of the horn tip is a low velocity pocket, suggesting the formation of a separation bubble within the shear layer attached to the upper surface, which is illustrated in **Figure 15**. The region reaches a peak turbulent viscosity ratio of 5860. The boundary layer subsequently reattaches at the upper surface of the trailing edge. The wake formed by the trailing edge is narrow; there are small pockets of recirculating air within the notches of the ice shape, however, this does not significantly influence the bulk flow.

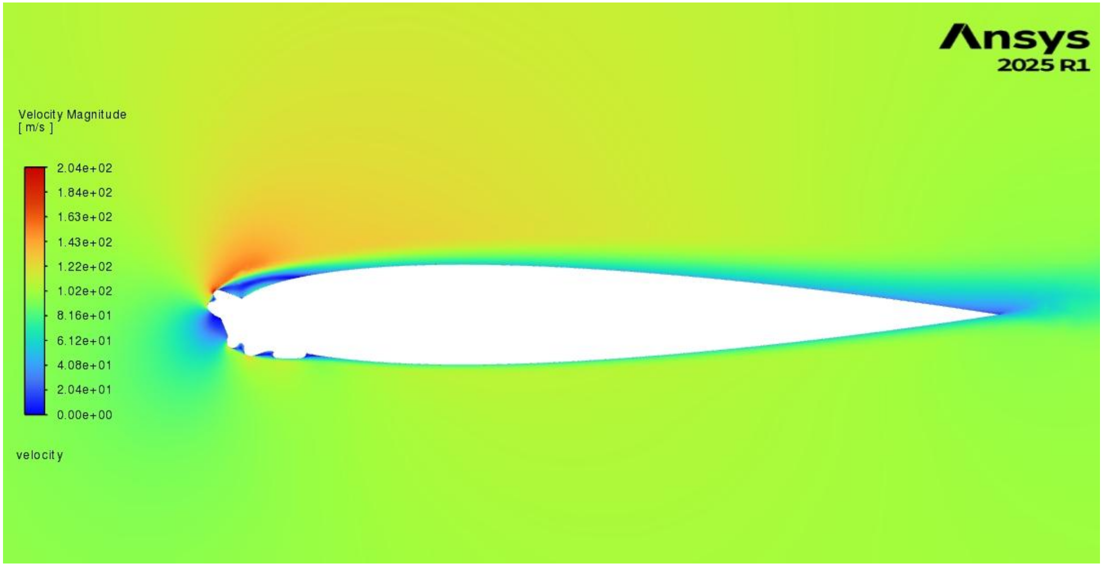

Figure 14: Velocity Magnitude contour for cruise angle of attack

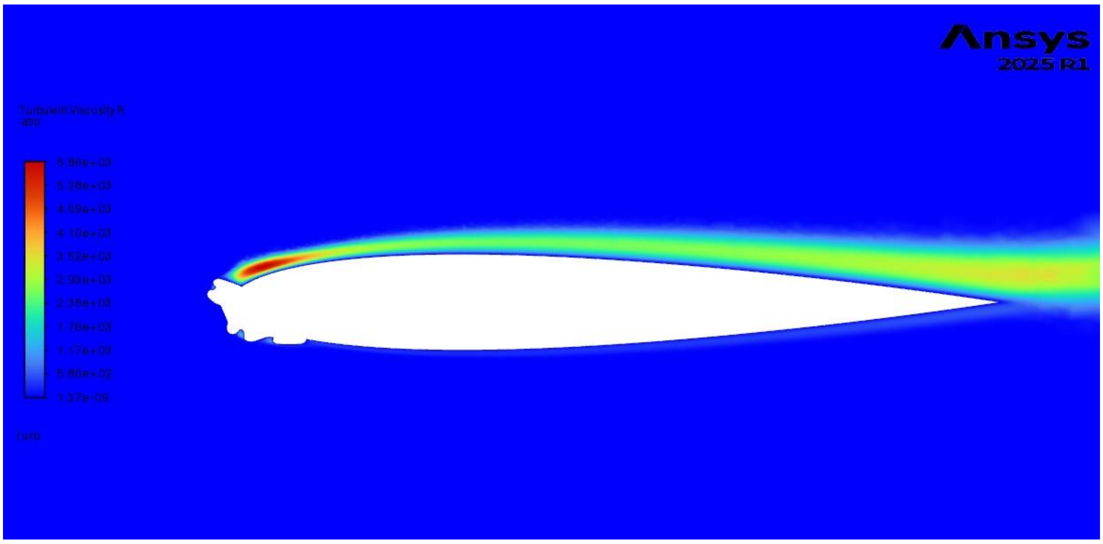

Figure 15: Turbulent Viscosity Ratio contour for cruise angle of attack

At the stall angle of attack, the stagnation point shifts down to the lower face of the glaze horn, intensifying the boundary layer acceleration at the upper surface, displayed in **Figure 16**. The separation bubble formed at the leading edge remains detached, whereas at the cruise angle it reattaches near the trailing edge. This forms a thick low velocity region, engulfing the entire upper surface, leading to a wide and slow wake. The peak turbulent viscosity ratio more than doubles from the value shown for the cruise angle of attack, achieving a peak value of 12,400, while simultaneously departing from the upper surface. The behaviour of the turbulent viscosity ratio is showcased in **Figure 17**. The shear layer that forms the boundary layer separates from the aerofoil beyond the mid chord region atop a low momentum recirculation zone spanning the aft portion of the upper surface. This shows that the separation bubble formed at the leading edge has failed to reattach to the aerofoil, merging with the adverse pressure gradient along the trailing edge of the aerofoil, suggesting that the upper surface is unable to contribute to the lift of the aerofoil. Thus, the aerofoil is presumed to have stalled.

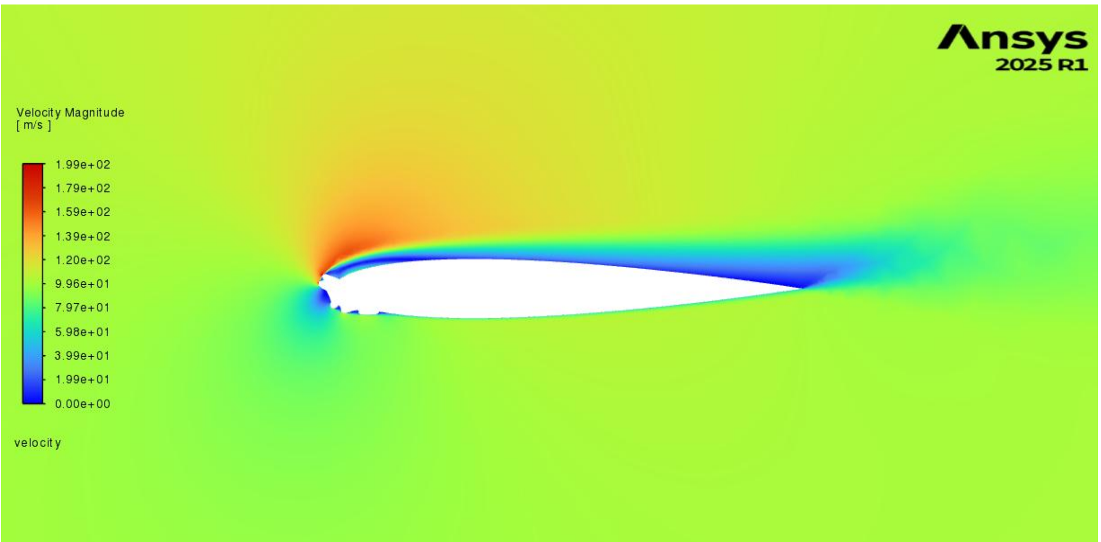

Figure 16: Velocity Magnitude contour for stall angle of attack

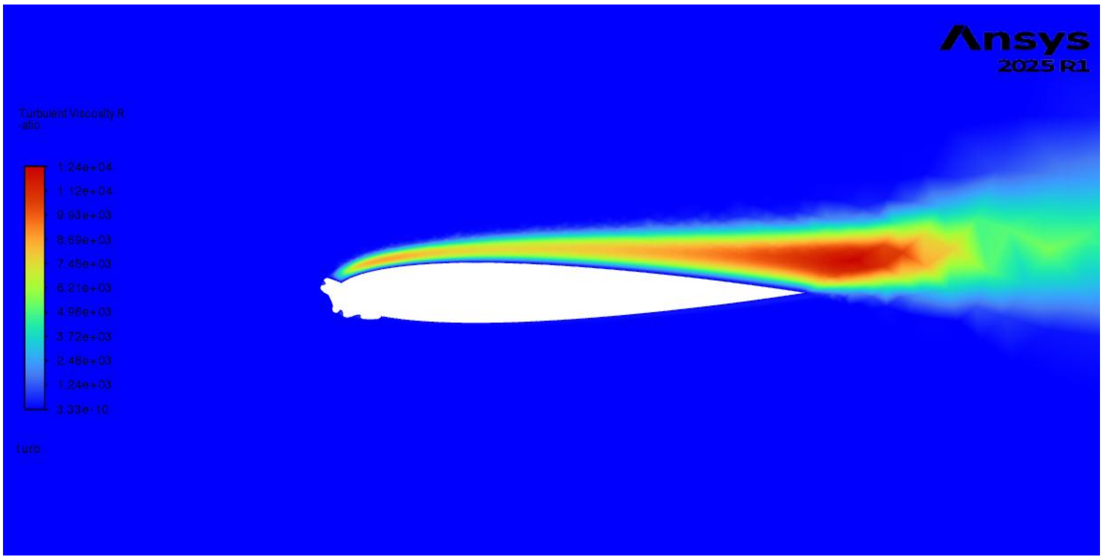

Figure 17: Turbulent Viscosity Ratio contour for stall angle of attack

The maximum lift coefficient of the iced aerofoil is 0.5574 at an angle of attack of 7°. This was derived by using the value of the last converged solution under second order discretisation. This is a 62% reduction in the Cl when compared to the clean aerofoil. Since convergence was diminished, with the force monitors displaying disorderly oscillating values, the true Cl for the iced configuration may yield a result that is slightly higher than 0.5574. The difference in Cl for angles of attack of 6° and 7° displayed a rise in the Cl of only 0.032, indicating the true coefficient of lift rise beyond 0.5574 would be minimal. The 62% reduction in the maximum Cl was documented as a lower bound for the true aerodynamic penalty.

The absolute values of Cl and Cd obtained are subjected to the limitations and numerical uncertainties typically encountered when conducting a fully turbulent Spalart Allmaras simulation on a thin symmetric aerofoil. Transition modelling was not enabled, and with Spalart Allmaras producing small offsets for the Cl and Cd for 2D aerofoil analysis, the Cl and Cd determined should be interpreted as the relative aerodynamic degradation rather than the absolute aerodynamic degradation. These engineering figures are the basis for conveying the findings of the analysis to the design team (WP1).

As the aircraft and iced shape were simplified to ensure reliability of the mesh, the drag coefficient increase is a lower bound on the actual drag coefficient increase. As the glaze horn tips were smoothed and the rime feathers were removed from the iced shape, the pressure drag was reduced relative to the true ice shape shown in the literature. Furthermore, no model for the surface roughness was applied to the iced wing. As mentioned in **Section 3.3,** the mismatching collection efficiency coupled with the employment of a y+ resolve to wall meshing strategy led to an unattainable accurate roughness constant. Thus, the drag coefficient figures reported only represent the smooth wall drag of the horn geometry.

The reduction in maximum Cl for the iced aerofoil results in a higher stall speed of the iced aircraft. The increase in stall velocity is determined using **Equation 28**:

$$V_{s,iced} = V_{s,clean} \times \sqrt{\frac{C_{L,max,clean}}{C_{L,max,iced}}} \tag{28}$$

By manipulating the general stall equation [49], the updated stall velocity can be determined. The design team has provided the stall velocity value of the aircraft, which is 61 knots. The stall velocity of the iced aerofoil was calculated to be approximately 99 knots, highlighting a 62% increase. Based on the obtained iced stall velocity, the holding speed margins for the OSCAR aircraft were advised to be at least 1.65 times the stall speed of the clean aerofoil.

4.4 Anti/De-Icing Technology Comparative Assessment

The icing technologies examined in the literature review were analysed using the methodology described in **Section 3.4**. The data collected from the literature review was arranged into three matrices, each reviewing one category of icing technologies (ice detection systems, active ice protection systems and passive ice protection systems). Within each matrix, the technologies were evaluated against the six criteria established within **Section 3.4**, with the assessment being performed and presented within this paper. An assessment of further technologies was performed and published within the comprehensive deliverable spreadsheet.

The first technology examined was the traditional thin etched-foil electrothermal system. These systems have a proven history of achieving certification within the aviation industry, and are compatible with CFRP when the heating elements are incorporated during the layup of the CFRP laminate [27]. However, an energy consumption analysis demonstrated that they are not a viable candidate for the OSCAR aircraft, as the high energy consumption makes it difficult to satisfy SC-VTOL.2430 [21]. These systems were therefore discounted as a viable standalone architecture for the OSCAR aircraft.

Systems utilising carbon nanotube (CNT) webs have been developed, with the aim of reducing the energy consumption of such systems. Additionally, the CNT webs have a density of 0.019 g/m^2 [29], making the weight contribution negligible [29]. CFRP material that includes the CNT webs retains the characteristics of standard CFRP material [29], and thus, fulfils the requirements within F3120/F3120M-20. However, there is a relative lack of certification experience for systems utilising CNT webs. While research has shown testing with CNT-based active ice protection systems (IPS), there is a lack of service history facilitating FIKI certification.

Technologies like electromechanical expulsion de-icing systems (EMEDS) suffer from high energy consumption. However, Cox & Co have obtained FAA Part 23 FIKI, facilitating the use of an EMEDS system as part of a hybrid system, which includes the use of a thermal parting strip [31]. However, two disadvantages limit EMEDS for the OSCAR system, and eVTOL use in general. Residual ice that remains between actuation cycles can influence ice build-up during operation. The shedding of the ice from the leading edge of the wing under SC-VTOL.2415 can be considered especially problematic for an eVTOL aircraft that includes a wing-mounted rotor [21].

The hybrid system known as ICE-WIPS, developed by JAXA and NASA involves the combination of an intermittent electrothermal heater at the leading edge of the aerofoil, and an icephobic coating, applied aft of the heated region [38]. Published IWT testing reports a 70% reduction in power requirements under Appendix C conditions, and a 30% reduction under conditions of high liquid water content (LWC) cumuliform conditions [38], when compared to traditional electrothermal systems.

The 30% power reduction is the more significant figure for the OSCAR aircraft, as the worst-case scenario, IM2, identified in **Section 4.2** is a high LWC cumuliform condition. Additionally, the system provides a solution to the runback ice; with the application of an icephobic coating aft of the heated region eliminating runback ice from refreezing on the unheated aerofoil surface [38]. This minimises the development of the extended ice runback zone for the IM2 condition, reducing the aerodynamic degradation.

When examining passive ice protection systems, the primary area of focus was to review systems using SLIPS and low surface energy coatings. SLIPS coatings exhibit ice adhesion strengths below 9 kPa [35], and are shown to reduce the weight of ice accumulating on the test substrates by up to 64% [34]. Testing of such systems, however, has revealed that the liquid treatment degrades in exposure to heavy rain, thereby limiting the use of SLIPS to surfaces experiencing low exposure.

Low surface energy coatings based upon POSS and nanosilica additives, which are used within a polyurethane based topcoat (Aviox 77702), have been shown to reduce glaze ice accretion by 65%, when tested on a NACA 0012 aerofoil. This provides direct results applicable to the OSCAR aircraft. Alternatively, slippery polyurethane (SPU) coatings have stable ice adhesion values of approximately 220 kPa, and have been tested against exposure to rain at rates of 175 m/s [39]. This trade-off of ice adhesion strength for greater usability in high-speed operation makes SPU the more durable option, where the leading edge of the aerofoil undergoes the greatest rain exposure.

Therefore, the type of coating recommended for the OSCAR aircraft varies according to the location of application. For instance, at the leading edge of the aerofoil, which is covered by the active ice protection system, an SPU or a modified polyurethane-based coating can be employed. This is due partly to the necessity to minimise or eliminate the formation of runback ice, which can otherwise result in severe aerodynamic penalties.

The two ice detection systems examined within the literature review were the fibre optic sensor arrays, and performance degradation prediction software. Fibre optic sensor systems can determine ice formation thickness on the aircraft in real time, with an accuracy of up to 10 mm [40]. Additionally, through spectral analysis, they can also determine the type of ice that forms on the leading edge [40]. This system can carry drawbacks, however, such as the requirement for installation of sealed, transparent windows onto the CFRP structure. This introduces manufacturing complexities, which can lead to high production costs, and a small weight penalty.

The second type of ice detection system is the performance degradation prediction software, which is shown to yield the greatest performance, alongside cost-effectiveness. Through the use of pre-recorded sensor data outlining the aircraft performance, it imposes no additional weight, while simultaneously fulfilling the requirements for primary detection within F3120/F3120M-20 Annex A4 [41].

Magnetostrictive resonant frequency probes, sold and deployed by companies like Collins Aerospace, have not been extensively covered in the literature review due to content constraints [50]. The technology was nevertheless assessed in the deliverables spreadsheet, which is made available to all OSCAR candidates.

For active ice protection, the ICE-WIPS architecture was recommended, as this system has been proven to satisfy the requirements within VTOL.2415 and VTOL.2430. For passive coating systems, a polyurethane-based coating is recommended for application to the leading edge, while a SLIPS or low surface energy based system was suggested for the aft of the wing and at the fuselage. Regarding ice detection systems, performance degradation prediction software was chosen, due to its minimal cost and weight constraints.

In reviewing the icing technologies, two main limitations were noted. Firstly, the validation of each of the technologies was based upon tests that were performed on fixed-wing aircraft that are covered within Part 23 of the FAA regulations. While the physics of such systems remain the same, the scenarios within the SC-VTOL regulation which relate to icing on rotor blades and during transitions between vertical and forward flight, have not been tested on aircraft similar to OSCAR. The second limitation is that the available data does not cover all of the coating assessment criteria, with some articles only publishing data for ice adhesion and contact angle, and others reporting data for durability and glaze ice reduction. Therefore, the decision on which coating to recommend was based on the available data for each coating on each criterion, rather than on data for each of the six criteria for each of the investigated coatings.

4.5 OSCAR Traceability

The deliverables produced within this project are not isolated outputs. Each deliverable influences the aims of other work packages throughout the OSCAR program. **Table 9** summarises the traceability relationships between the work packages during this project.

ASTM F3120/F3120M-20 and EUROCAE ED-314 were used to construct the AFM framework, and to include all necessary flight procedures, to contribute to the creation of the pilot training program. The anti/de-icing technology comparative assessment was uploaded to the OSCAR database, and serves as a guide for the design team (WP1), providing a list of technologies that can achieve certification, ensure safe operation and mitigate the aerodynamic degradation experienced when flying in critical glaze icing conditions. When applicable, the listed manufacturability can aid the manufacturing team (WP2) in developing manufacturing procedures, based on the selection of technologies made by the design team. The certification compliance matrix for Appendix C icing serves as a microcosm of the overall aim of WP4, which is to ensure the aircraft is certified under the relevant aviation authorities.

Table 9: Project Deliverables Traceability Matrix

Requirement(s):	Input(s):	Project Deliverable(s):	Output(s):
Determine Aerodynamic Degradation caused by worst-case Appendix C icing	**WP1 -** OSCAR wing geometry and aircraft velocities.	Case Study: Aerodynamic Degradation for worst-case Appendix C	**WP1 –** Aerodynamic degradation and stall speeds. **WP3 –** Aircraft holding speed (Under severe glaze ice)
Create a comprehensive guide on anti/de-icing technologies	Existing Literature	Comparative Assessment Matrices	**WP1 –** Guide for how to tackle aircraft icing **WP2 –** Manufacturing descriptions (If technology is adopted by **WP1)**
Provide relevant icing contribution in the AFM & Sample AFM for icing	ASTM F3120/F3120M-20 & EUROCAE ED-314	AFM Sample Report for icing	**WP3 –** Guide for pilot training when flying in icing & icing requirements
Construct Comprehensive Certification Matrix for Appendix C Icing	**WP0** - Region of aircraft deployment	Certification Compliance Matrix with Testing Methodologies	**All WP's** – Ensures aircraft is certified, provides requirements and testing methods.

5.0 Conclusion

A comprehensive certification, compliance, and testing matrix was established for urban air mobility platforms, specifically an eVTOL operating under EASA Appendix C icing conditions. The certification matrix provides guidance for achieving compliance and testing strategies that demonstrate compliance. The matrix consolidates the icing requirements of four specification regulatory frameworks (CS-23 Amendment 6, SC-VTOL Issue 2, CS-P Amendment 2 and SC E-19 Issue 1). This process highlighted 13 top-level certification requirements, which are linked to 110 MOC sub-requirements across five external standards. Each of the sub-requirements specifies one or more testing methodologies, which allows for the consolidation of testing campaigns to reduce the certification cost and time for the OSCAR programme.

The case study formulated a structured methodology to estimate the impact of critical ice accretion on aircraft performance, without the use of a specialised icing CFD software, such as ANSYS FENSAP-ICE. Four Appendix C conditions were analysed, favouring higher LWC to identify the most hazardous icing scenario, at a static temperature of -8.5°C. The similarity parameters for the worst-case scenario icing, for both continuous maximum and intermittent icing, were determined. The continuous maximum condition was paired with Anderson's Run 414 ice shape. There was, however, no applicable literature ice shape matching the similarity parameters of the worst-case intermittent maximum conditions.

The literature ice shape was sketched onto the OSCAR aerofoil, and underwent CFD simulation. Analysis confirmed substantial degradation in the lift and an increase in drag. At the cruise angle of attack, the iced aerofoil experienced a 14% reduction in the coefficient of lift and a 169% increase in the coefficient of drag, when compared to the clean aerofoil. The effective stall angle of attack for the iced aerofoil was deemed to be 7°, a 9° reduction from the clean aerofoil's stall angle. The maximum lift coefficient of the iced aerofoil showed a 61.7% reduction, translating into a stall velocity increase from 61 knots to 99 knots. The deliverables produced serve as an integrated pre-certification plan. The certification matrix provides a regulatory overview, the case study determines the aerodynamic threat, and the IPS recommendation offers an icing mitigation strategy. The outputs produced and the degradation estimates were provided to the design team, allowing for consideration of the parameters when designing the aircraft.

An extensive literature review on active and passive anti/de-icing technologies was performed to serve as a comparative guide, ensuring the selection of IPS aligns with the specific aircraft configuration. The review of passive anti/de-icing technologies also aimed to mitigate the threat of icing itself, through the reduction of adhesion and accumulation of ice on the aircraft surface, easing the operational strain on the active protection system. Hybrid systems were assessed as the most viable for standard eVTOL aircraft, with the recommended architecture consisting of the ICE-WIPS hybrid system coupled with polyurethane and low surface energy coating applied across different regions on the aircraft. Performance degradation prediction software was determined as the optimal ice detection system for the OSCAR aircraft.

Future work should prioritise expanding the ice severity analysis to include propeller icing degradation, as this aids in determining the thrust loss, and is required in ASTM F3120/F3120M-20 Annex A1 [19]. Acquiring access to EUROCAE ED-103A and SAE AS5562 can refine the certification matrix, providing detailed operational performance requirements. Additionally, obtaining accurate performance degradation data for intermittent maximum worst-case is vital, which can be done through specialised icing CFD software such as ANSYS FENSAP-ICE. Finally, the certification compliance matrix should be expanded to incorporate cost and time estimates, enabling the creation of a cost and time effective certification plan.

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By submitting this form as part of an assessment, you confirm that your disclosed usage of generative AI tools is accurately accounted for. Disclosure through this form does not absolve you of the requirement to follow the university's policies and procedures surrounding academic honesty and integrity. Please see the university's policies and procedures in addition to your department's undergraduate handbook for more specific guidance on the use of generative AI.

What generative AI tools did you use?	ChatGPT, Claude
Explain the reasoning for your use of these tools.	Drafting a Nomenclature and Appendix B. I used these because I wished to save time. Making a nomenclature and ensuring the units are consistent is an extremely time consuming process which can be done quickly by AI. Inputting equations in constantly into the Appendix is a time consuming process which can be made much simpler and quicker with the help of AI.
Describe how you used it/them. You may wish to include a selection of representative prompts or links to your chats.	I used it to construct a nomenclature for my project, ensuring all abbreviation were clearly defined with its units explained. For Claude, I inputted my written equations into it asking it could make a presentable table for me with all the equations inside.
Identify any risks introduced by your use of AI and how you mitigated these risks. Describe your process of quality assurance.	There was a risk of the AI missing an abbreviation. I proof read the document, ensuring all abbreviations were explained. I ensured that no formatting errors or misplaced characters were present. For the Appendix, there was a threat of the calculations appearing wrong or the formula being written incorrectly. I have an excel sheet for my scaling calculations, I used that to verify and validate the values.

7.0 Appendices

7.1 Appendix A: Certification Compliance Matrix for Icing

This appendix presents the complete two-layer certification compliance matrix discussed in Section 4.1. The upper layer maps the 17 icing-relevant top-level requirements which are extracted from the four certification frameworks (CS-23 Amendment 6, SC-VTOL Issue 2, CS-P Amendment 2, and SC E-19 Issue 1) and links to their applicable means of compliance documents.

The data is the means of compliance column is a hyperlink which will only work within the excel spreadsheet and my computer. This is shown in **Table 10**.

Table 10: [APPENDIX A] Upper Layer Certification Table

Certification Specification:	Certification Version:	Certification Requirement:	Description of Requirement:	Method of Compliance Document (MOC):
CS - 23	Amendment 6; AMC/GM Issue 4	**CS 23.2165:** Performance and flight characteristics requirements for flight in icing conditions	An applicant who requests certification for flight in icing conditions must demonstrate a means to detect any icing condition for which certification is sought after, along with a system that provides safety warnings and limitations for the pilot to comply with.	F3120M-20:Standard Specification for Ice Protection for General Aviation Aircraft
CS - 23	Amendment 6; AMC/GM Issue 4	**CS 23.2415:** Powerplant Installation Ice Protection	The airplane design and powerplant installation design must prevent predicted ice accretion and ice shedding that would harm the powerplant operation within the icing conditions for which certification is requested	F3120M-20:Standard Specification for Ice Protection for General Aviation Aircraft
CS - 23	Amendment 6; AMC/GM Issue 4	**CS 23.2415:** Powerplant Installation Ice Protection	The airplane design and powerplant installation design must prevent predicted ice accretion and ice shedding that would harm the powerplant	F3062/F3062M-20: Standard Specification for Aircraft Powerplant Installation

			operation within the icing conditions for which certification is requested	
CS - 23	Amendment 6; AMC/GM Issue 4	**CS 23.2415:** Powerplant Installation Ice Protection	The airplane design and powerplant installation design must prevent predicted ice accretion and ice shedding that would harm the powerplant operation within the icing conditions for which certification is requested	F3063/F3063M -20: Standard Specification for Aircraft Fuel Storage and Delivery
CS - 23	Amendment 6; AMC/GM Issue 4	**CS 23.2415:** Powerplant Installation Ice Protection	The airplane design and powerplant installation design must prevent predicted ice accretion and ice shedding that would harm the powerplant operation within the icing conditions for which certification is requested	F3066/F3066M -18: Standard Specification for Aircraft Powerplant Installation Hazard Mitigation
CS - 23	Amendment 6; AMC/GM Issue 4	**CS 23.2540**: Flight in Icing Conditions	To achieve certification for FIKI conditions, the ice protection system must allow for safe operation and the aircraft design must provide protection from stalling during autopilot operation.	F3120M-20:Standard Specification for Ice Protection for General Aviation Aircraft & F3233/F3233M -21: Standard Specification for Flight and Navigation Instrumentation in Aircraft & F3061/F3061M -20: Standard Specification for Systems and Equipment in Small Aircraft

SC - VTOL	Issue 2	**VTOL.2165:** Flight in Icing Conditions	Aircraft must be able to safely operate withing the requested icing conditions. An icing detection system must be employed to avoid encountering icing conditions the plane is not certified for. Operating limitations must be developed to prohibit intentional flight, including take-off and landing, into icing conditions for which the aircraft is not certified for	EUROCAE ED-314: Compliance methodologies for VTOL certification in inadvertent icing and snow operation
SC - VTOL	Issue 2	**VTOL.2415:** Lift/Thrust system installation ice protection	The aircraft and lift/thrust system must be designed with ice shedding precautions considerate to ensure safe system operation.	EUROCAE ED-314: Compliance methodologies for VTOL certification in inadvertent icing and snow operation
SC E-19	Issue 1	**EHPS.80:** Safety Assessment	An analysis of the EHPS, including the control system, must be carried out to assess all possible failure conditions that can reasonably be expected to occur. This should be done for all control surfaces such as icing detectors and ice protection systems. Probability of failure occuring must be assessed along with a summary made of possible failing conditions that can result in hazardous and catastrophic effects. Operating instructions and contingency plans/systems must be in place to minimise hazards.	F3120M-20:Standard Specification for Ice Protection for General Aviation Aircraft

SC E-19	Issue 1	**EHPS.280:** Icing and Snow Conditions	The electric/hybrid propulsion system and other of the sub components must be able to perform safely and satisfactorily when operating in icing conditions. Any inlets and ducts must not be blocked when operating in icing conditions	EUROCAE ED-314: Compliance methodologies for VTOL certification in inadvertent icing and snow operation
SC E-19	Issue 1	**EHPS.280:** Icing and Snow Conditions	The electric/hybrid propulsion system and other of the sub components must be able to perform safely and satisfactorily when operating in icing conditions. Any inlets and ducts must not be blocked when operating in icing conditions	F3120M-20:Standard Specification for Ice Protection for General Aviation Aircraft
SC E-19	Issue 1	**EHPS.300:** Fuel System	If the EHPS posses a reciprocating engine or a turbine engine, evidence must be provided that demonstrates the engine fuel system capable of withstanding flight conditions its being certified for.	F3063/F3063M-20: Standard Specification for Aircraft Fuel Storage and Delivery
SC E-19	Issue 1	**EHPS.350:** EHPS Control System	The aircraft's control system must be designed/implemented so that the EHPS operates as intended without alarming power oscillations. There must be no difficulties in starting and utilising the control system, software must be developed using a structured and methodical approach that provides a level of assurance in the event of an unexpected hazard. Must be able to comply with EHPS.460 Operational Demonstrations.	F3120M-20:Standard Specification for Ice Protection for General Aviation Aircraft

SC E-19	Issue 1	**EHPS.460:** Operational Demonstration	The aircraft must be able to reliabily start, shut down, restart and operate within the certified flight conditions (FIKI).	F3120M-20:Standard Specification for Ice Protection for General Aviation Aircraft
CS-P	Amendment 2	**CS-P 350c:** Centrifugal Load Tests	De-icing equipment that is to be used or attached to the propellor must be capable of withstanding a load equivalent to 150% of the maximum centrifugal load in which the component would be subjected at the maximum permissible/governed rotational speed for 30 minutes. This can be through analysis if plausible.	AMC P 350c Centrifugal Load Tests
CS-P	Amendment 2	**CS-P 380:** Lightning Strike	Ensure the de-icing system is able to withstand a lightning strike and operate as intended.	AMC P 380 Lightning Strike
CS-P	Amendment 2	**CS-P 390:** Endurance Testing	Ensure the de-icing system is installed during the endurance tested. De-icing components must not degrade the propellers performance when used for prolonged periods of time	AMC P 390 Endurance Testing

The lower layer consists of the 110 MOC sub-requirements derived from the five external MOC standards (ASTM F3120/F3120M-20, EUROCAE ED-314, ASTM F3316/F3316M-19, ASTM F3062/F3062M-20, ASTM F3063/F3063M-20 ,ASTM F3066/F3066M-18 and CS-P) each paired with one or more applicable testing methodologies. The matrix is organised by MOC document to reflect how it is navigated during a certification campaign.

Due to Copyright Reasons. ALL MOC such as ASTM F3120/F3120M-20 are unable to be shown. 'WARNING: ASTM DOCUMENTATION AND TESTING MATRIX MUST ONLY BE USED FOR INTERNAL COMPLIANCE PURPOSES ONLY. REDISTRUBITION OR COMMERCIALISATION OF CONTENT IS STRICTLY PROHIBITED!'

7.2 Appendix B: Icing Case Study

This appendix presents the full numerical calculations performed for all four Appendix C conditions sampled in Section 4.2: two continuous maximum conditions (CM1 and CM2) and two intermittent maximum conditions (IM1 and IM2). The calculation chain follows the methodology established in Section 3.2, with each parameter presented alongside its governing equation and its computed value for each condition. All four cases share the same altitude (10,000 ft), static temperature (−8.5 °C), and characteristic dimension (d = 0.07196 m); they differ only in their droplet size (MVD) and liquid water content (LWC), as tabulated below.

Table 11: [APPENDIX B] Shared Input Parameters

Parameter	Value
Aerofoil section	NACA 0012
Mean aerodynamic chord (MAC)	2.2773 m
MAC with sweep correction	2.3990 m
Sweep angle (Λ)	18.33°
Analysis altitude	10,000 ft
Static temperature ($T\infty$)	−8.5 °C
Holding airspeed (indicated) with + 10 buffer (Certification)	200 kts
Holding airspeed (true) + 10 buffer (Certification)	102.89 m/s
Exposure time	1,350 s
Characteristic dimension (d = 2r_LE)	0.07196 m

Table 12: [APPENDIX B] Case-Specific Input Diameter

Parameter	CM1	CM2	IM1	IM2
Cloud type	Stratiform	Stratiform	Cumuliform	Cumuliform
Mean volumetric droplet diameter (MVD)	15 µm	25 µm	15 µm	20 µm
Liquid water content (LWC)	0.6300 g/m³	0.3347 g/m³	2.5585 g/m³	2.1984 g/m³

Table 13: [APPENDIX B] Atmospheric and Fluid Properties (Shared with all Cases)

Parameter	Equation / definition	Value
Static pressure	$P_\infty = P_0 (1 - L \cdot h / T_0)^{(g/LR)}$	69,676.84 Pa
Air density at altitude	$\rho_\infty = P_\infty / (R \cdot T_\infty)$	0.9173 kg/m³
Film temperature	$T_f = (T_\infty + T_s) / 2$	268.9 K
Air density at film T	$\rho_f = P_\infty / (R \cdot T_f)$	0.9029 kg/m³
Air viscosity at static T	$\mu_\infty = \mu_0 (T_\infty/T_0)^{1.5} (T_0+S)/(T_\infty+S)$	1.6736×10^{-5} Pa·s
Air viscosity at film T	μ_f (Sutherland's law at T_f)	1.6949×10^{-5} Pa·s
Thermal conductivity at film T	$k_f = c_p \mu_f / Pr_f$	0.02392 W/m·K
Prandtl number at film T	$Pr_f = c_p \mu_f / k_f$	0.7120
Water vapour diffusivity	$D_v = 2.11 \times 10^{-5} (T_f/273.15)^{1.94} (101325/P_\infty)$	2.9764×10^{-5} m²/s
Schmidt number at film T	$Sc_f = \mu_f / (\rho_f D_v)$	0.6307
Reynolds number at film T	$Re_f = \rho_f V d / \mu_f$	394,419
Nusselt number	$Nu = 1.14 Re_f^{0.5} Pr_f^{0.4}$	624.99
Convective heat transfer coefficient	$h_c = Nu k_f / d$	207.78 W/m²·K
Mass transfer coefficient	$h_m = h_c Sc_f / (c_p Pr_f)$	0.2242 kg/m²·s
Vapour pressure at surface	p_{vw} (at $T_s = 0$ °C)	610.78 Pa
Vapour pressure at static T	$p_{v\infty}$ (at T_∞)	322.13 Pa
Recovery factor	$r = Pr_f^{(1/3)}$	1.00

Table 14: [APPENDIX B] Similarity Parameter Calculations

Parameter	Equation	CM1	CM2	IM1	IM2
Inertia parameter (K)	$K = \rho_w MVD^2 V / (18 d \mu_\infty)$	1.0646	2.9573	1.0646	1.8927
Droplet Reynolds number (Re_d)	$Re_d = \rho_\infty MVD V / \mu_\infty$	84.59	140.99	84.59	112.79
Stokes ratio (λ/λ_s)	$\lambda/\lambda_s = 1 / (1 + 0.197 Re_d^{0.63} + 2.6 \times 10^{-4} Re_d^{1.38})$	0.3755	0.3085	0.3755	0.3370

Parameter	Equation	CM1	CM2	IM1	IM2
Modified inertia parameter (K_0)	$K_0 = (1/8) + (K - 1/8)(\lambda/\lambda\_s)$	0.4778	0.9989	0.4778	0.7207
Collection efficiency (β_0)	$\beta_0 = 1.40 (K_0 - 1/8)$^0.84 $\cos^2\Lambda$	0.3498	0.5556	0.3498	0.4512
Accumulation parameter (Ac)	Ac = LWC·V·τ / (ρ_ice d)	1.3260	0.7046	5.3852	4.6272
Relative heat factor (b)	b = LWC V β_0 c_p,w / h_c	0.4602	0.369	1.869	2.07
Water energy transfer (φ)	φ = (T_s – T∞) – V^2 / (2 c_p,w)	7.24	7.24	7.245	7.245
Air energy transfer (θ)	θ = (T_s – T∞) + (h_m/h_c)[(p_vw–p_v∞) L_v / (P∞ c_p)]	14.41	14.41	14.4	14.4
Weber number (δ)	δ = ρ∞ V^2 d / σ	2,435.6	4,059.4	2,435.6	3,247.5
Weber number (Weber_c)	Weber_c = ρ_w V^2 d / σ	10,752	10,752	10,752	10,752
Weber number (Weber_l)	Weber_l = ρ_w V^2 MVD / σ	1.17×10^7	1.17×10^7	1.17×10^7	1.17×10^7
Freezing fraction (n)	n = (c_p,w / L_f) (φ + θ/b)	0.4867	0.5598	0.1888	0.1793

Table 15: [APPENDIX B] Outputs for Ice Geometry

Parameter	Equation	CM1	CM2	IM1	IM2
Leading edge ice thickness	h_LE = Ac d (ρ_w / ρ_ice)	16.25 mm	15.77 mm	25.59 mm	26.94 mm
Runback zone distance	h_runback = h_LE,rime – h_LE,glaze	17.14 mm	12.40 mm	109.98 mm	123.32 mm

The worst-case condition selected for CFD analysis was CM1 (continuous maximum, MVD = 15 μm, LWC = 0.63 g/m³), paired with Anderson's Run 414 reference ice shape. The worst-case intermittent maximum condition was IM2 (MVD = 20 μm, LWC = 2.20 g/m³), which produced the lowest freezing fraction (0.1793) and the largest runback zone (123.32 mm) of the four cases. Selection rationale is detailed in Section 4.2.

7.3 Appendix C: Comparative Anti/De-icing Technologies Assessment

This appendix presents the complete technology matrices compiled during the comparative assessment described in Section 4.4. The scope extends beyond the technologies discussed in the main text to cover the broader survey of technologies reviewed during the project. The matrices are organised into three categories following the structure established in **Section 2** and **3.4.**

1. Ice detection systems
2. Active and Hybrid ice protection systems
3. Passive coatings.

Each entry is assessed against the 6 main criteria, however, due to limited data coverage, only partial fields have been fulfilled. The matrices serve as a guide for the OSCAR design team in tackling icing. One thing to note the imported tables DO NOT contain the source link which is in the excel document, formatting errors made this difficult to facilitate

Table 16: [APPENDIX C] Ice Detection System Matrix

Name	**Features**	**Results**	**Certification Alignment**
Fiber Optic Array Ice Sensors	Involves embedding fibre optics into the wing. It measures light that reflects off any present ice. By recording how the light behaves , the thickness and type of ice and can be determined.	Can accurately measure ice thickness (up to 10mm) and identify the type of ice in real time. However, a sealed transparent glass protection window must be built directly into the leading edge of the aircraft, making installation complex. Blind spots can also arise if ice is shown to be extremely thin. This sensor was explicitly tested on a vertical NACA0012 wing.	Must meet the rules outlined in ASTM F3120 (Section 9.1.9 and Annex A4). Must pass EUROCAE ED-103A or SAE AS5498A response time tests
Flexible Ultra-Thin Ultrasonic Transducer	An extremely thin flexible sensor patch that measures ice thickness by sending out ultrasonic pulses and listening to the return signal.	Extremely lightweight and bends easily around curved surfaces. It is sensitive, detects ice as thin as 0.29mm. However, a watery film on top of the ice can result in signal interference, along with inaccurate thickness values (usually underestimated).	Can be used for curved surfaces and external components. However will need to qualify through passing EUROCAE ED-103A threshold standard tests.

		However, this sensor was tested with aluminium, CFRP has a lower acoustic impedance, this would require an alternative approach to prevent signal oscillation.	
UAV Based Spectral Ice Detection System	To be used for preflight checks. A drone flies around the aircraft during ground turnover, taking pictures with a camera that captures different light wavelengths. Artificial Intelligence then scans the photos to spot ice.	Allows for safe and automated preflight check. It's main use is in detecting clear ice that may be hard to see. It achieves an overall detection accuracy of 88-92% and glassy clear ice detection up to 97%. However, processing takes 17 minutes, which is rather lengthy if turnover times are short. Processing time can be reduced however, accuracy may decrease as a result.	Fulfils the mandatory physical ground inspection requirements for the wing leading edge while providing a safer alternative to climbing on the aircraft for pre take off wing checks (avoids high voltage risk areas).
Graphene Based AI Ice Control System	This a lightweight thin layer of graphene built into the aircrafts leading edge skin. This passively detects the instant heat release that occurs when water first freezes (latent heat of freezing) through the use of AI. This lets the system known to activate and melt the ice.	This technology was validated by embedding graphene strips between GFRP (Graphene) layers, this shows the installation process remains the same when installing into CFRP. This sensor acts as a 0 power passive sensor, detecting the latent heat released when SWD freezes. The machine learning model achieved a 60-62% accuracy in predicting ice formation through temperature spikes. Since graphene has high thermal and electrical conductivity, the layers heat up instantly and melt the ice.	Fulfils the criteria for an automated primary ice detection system. AI must be trained and tested to avoid nonsensical activations at non icing speeds.

Performance Degradation Prediction Software	A software continuously calculates the aircraft's expected nominal performance (Expected thrust, drag, and power consumption) using internal physics and propulsion models. Baseline/clean aircraft performance is compared against real time data gathered from the aircrafts standard sensor profiles (GNSS, IMU, PITOT TUB, TAS), advanced estimation algorithms such as Extended Kalman Filters are used to analyse the residuals and assess possible ice accretion (based on degradation).	There is virtually no financial cost or physical weight added, this makes it ideal to implement for VTOL with size/power/weight constraints. Can also be implement in conjunction with other detection technologies to automate IPS activation. Main con is that this requires a lot of computational resources.	Satisfies the ASTM F3120 Annex A4 requirements as there is no addition of redundant heavy hardware. IPS activation must be annunciated along with any critical performance values observed.
Infared Thermal Sensors	System consists of an external active thermal excitation source paired with an infrared camera. The heater or whatever heat source is equipped emits a short pulse of heat to irradiate the surface, the infrared imager captures the thermal radiation as the surface absorbs the heat and cools down. Analysis of the heat transfer rate allows the system to determine the presence of liquid water or ice, the shape/profile of the ice can be determined if the data is ran through edge detection methods.	System reconstructs 3D ice shapes and estimates ice thickness with less than a 2% margin error, it also is able to differentiate between water and ice. However, equipment costs are expensive as well as installation being complex.	Aids in achieving compliance with ASTM F3120 Annex A4.3 (nuisance activations).
Resonant Frequency Probes	This involves the use of magneto-restrictive coils to continuously vibrate a ferromagnetic probe at its natural frequency. This allows for physical detection of ice accretion, the mass of the ice formation on the probe dampens the vibration, resulting in a change of	Utilised by Collins Aerospace, this serves as a smart trigger for the main IPS, reducing continuous heating operation time by 75%	Satisfies ANNEX A4, and Section 9.1.5 by preventing the heaters from wasting power.

	frequency. The system inputs this change in frequency to trigger an alert for possible icing.		

Table 17: [APPENDIX C] IPS System Matrix

Name	**Features**	**Results**	**Certification Alignment**
Thin Etched Foil Electrothermal Heating	Utilises metallic meshes/thin etched foils encapsulated within insulated polyimide layers. These metallic meshes are then co cured into the CFRP. This IPS works through the generation of heat via electrical resistance, this is referred to as Joule Heating, this prevents supercooled water droplets from freezing upon impact or melts the formed ice just enough that the adhesive strength lowers and the airflow can push it off.	Easily appliable to curved structures and is lightweight while requiring simple maintenance (no mechanical moving parts). However, this may would be difficult for use in an eVTOL as a result of its excessive energy demands (26.83kW/m^2 in turbulent flight conditions). If the metallic meshes are spaced out, unheated gaps arise which help in the formation of runback ice.	Would be difficult to satisfy ASTM F3120 9.1.5 due to the heavy power consumption.
Carbon Nanotube Webs	Carbon Nanotubes are manufactured into thin webs. When the aerofoil is being fabricated, these webs are integrated directly between the layers of the CFRP composite structure. These work by generating heat through electrical resistance (similar to thin etched foil). The layout of the nanotubes can influence the concentration of heat, allowing variation of heating profile.	Very little weight (density of only 0.019g/m^2), negligible weight penalty. Additionally, they operate under low voltages, making CNT's a power efficient solution. However, there are many precautions that must be taken when manufacturing. Slight variations in the physical pressure applied during the curing process can alter the composite's fibre volume fraction, which	Satisfies ASTM F3120 9.1.5 AND 9.1.4 as it retains the structural characteristics of un-modified CFRP as well as requiring very little power.

		can affect the systems heating capabilities.	
ICE WIPS	A hybrid system that merges an active electrical heating source with effective passive coatings to maximise efficiency (minimising power consumption). The aircraft is divided into specific zones, an electrothermal heater is installed at the leading edge/stagnation point and is layered with a durable protective paint to minimise erosion. Directly behind the heaters, a superhydrophobic coating is applied. The heater melts the ice to a point that allows the airflow to blow it off, a superhydrophobic coating is applied directly after to ensure the ice is fully repelled and to prevent the formation of runback ice.	This reduces total power consumption by up to 70% in standard appendix C icing conditions, and 30% in high LWC conditions. However, superhydrophobic currently suffer from low resistance to rain erosion. The coating will need to be reinforced to maximise its usage and effectiveness.	Personally, I'd say this technology (if paired with a erosion resistant coating) is the ideal technology for achieving certification. Low power consumption, prevention of runback ice and high overall efficiency make this the leading choice from a certification perspective.
Spatiotemporal Self Heating	Very little amount of the insulating surface resin is removed, copper electrodes are attached to the exposed carbon fibres, effectively transforming the aerofoils skin into an electrothermal heater. By positioning the electrodes at the leading edge of the wing, thermal energy can be concentrated at the stagnation point, maximising the effectiveness of the	Showcases high energy efficiency with negligible weight penalties. Power density is 0.42kW/m^2 for anti icing activity, and 0.99kW/m^2 for de icing activity. Heat dissipation losses reduce by up to 99% if system is utilised efficiently (through pulses rather than continuous usage). The main limitations with this technology is the high demand of monitoring	Satisfies ASTM F3120 9.1.5 due to its minimal power consumption. However, would be difficult to certify for section 9.1.4 due to its catastrophic risk level. This would make certifying this system a nightmare.

	heating system. Additionally, the amount of energy/heat generated can vary depending on the pilot's requirements, a continious DC current can maintain a baseline temperature or fast-high power electrical pulses can be transmitted to melt the formation of the ice (allowing for aerodynamic removal).	thermal readings, ensuring the composite does not overheat. If the composite overheats, the structural integrity of the components will be compromised. This makes usage of this technology high risk - high reward.	
Electro Mechanical Expulsion De-Icing System (EMEDS)	Utilises mechanical actuators like electromagnetic coils or ultrasonic wave generates to fracture and expel ice from the aircraft. The mechanical actuators are installed inside or underneath the skin of the aerofoil or rotor. When ice forms, the actuators generate a mechanical shockwave/excessive vibration that travels outward through the composite skin. This results in a reduction of ice adhesive strength, allowing the airstream to expel the ice. This system works similar to pneumatic boots.	Cox and Co have achieved FAA Certification for FIKI on Part 23 aircraft (when used as a hybrid, paired with a thermal parting strip). Energy consumption is low compared to thermal melting systems. With the skin of the aerofoil comprising of CFRP, structural fatigue and delamination risks are to be considered (through strong pulsating forces). Additionally, a thin layer of residual ice is usually left behind as the pulse is unable to shake off all of the formed ice (reinforcing the need of a thermal strip).	Ice shedding analysis will need to be undertaken to ensure ice will not damage critical components. The expulsion system needs to be tested on CFRP to ensure no structural degradation.

Table 18: [APPENDIX C] Windshield Coatings

Product Name	Manufacturer	Description	Modularity	Durability	Source
Surface Seal	PPG Aerospace	Keeps windshield clear in rain without the use of wipers, ensure pilot is able to see visual cues of ice accretion	Comes in the form of wipes and can be applied during turnover in 15 minutes.	Not Durable. This is a replenishable product and should be treated as a consumable rather than a permanent item.	https://ppgaerospacestore.com/products/surface-seal-kits?variant=41111116283964
NANO MYTE SuperAi	NEI Corp.	A nanocomposite that can be sprayed. Ideal use for non critical surfaces. Reduces ice adhesion by approx. 80%	Comes ready to use as a liquid. Can be brushed or sprayed on.	Rated for 1000+ hours weatherability. Lasted 6 months in auto tests with no signs of degradation	https://www.neicorporation.com/products/coatings/anti-ice-coatings/

Due to the sheer size of the tables, the passive coating matrix was imported via screenshots

Table 19: [Appendix C] Passive Coatings Matrix Part 1

Coating Strategy	Configuration/Method	Mechanism	Strengths	Weaknesses	Possible Applications	Certification Alignment	Source of Study
egrated superhydrophobic CFRP	Fabrication by hotpressing hydrophobic silica nanoparticles into the CFRP prepreg using a 1200 mesh sieve template. The nano structures can also be fabricated through Femtosecond Laser Direct Writing	It establishes a highly stable Cassie-Baxter state through insulated air pockets, minimising the solid liquid contact fraction. This results in an increase of the freezing time of supercooled droplets by more then 3.5 fold (78 secs --> 283s).	No risk of the coating peeling or delaminating sicne the topography is cured/etched into the composite itself. An increase in tensile strength is observed (fromt 289.9 MPa to 320.5 Mpa). Droplets fuly rebound of sources (when possessing Weber numbers of 20-140) when the surface temperature is above -25℃	In icing wind tunnels simulating 50m/s speeds, the dynamic pressure of the droplet exceeds 1 MPa, this forces the droplets to transition into the Wenzel State (Weak hydrophobicity). Indicating this coating is not effective for high speed dynamics.	Can perhaps be used on areas where droplet impact pressures are lower, such as rearward portions of the wing and fuselage	Can be used when exiting inadvertent icing conditions. A delay in the droplet freezing process provides a safety buffer, allowing the pilot to safely exit the icing cloud.	https://doi.org/10.1021/acsami.2c15253?urlappend=%3Fref%3DPDF&jav=VoR&rel=cite-as. https://doi.org/10.1021/acsapm.5c02689?urlappend=%3Fref%3DPDF&jav=VoR&rel=cite-as. https://doi.org/10.3390/nano12162782
Hybrid Active-Passive Electrothermal System	An active heater complemented with a superhydrophobic coating	The active heater is tasked with preventing instantaneous condensation freezing, while the superhydrophobic coating acts as a thermal insulator against convective heat loss, drastically lowering the ice adhesion.	Prevents the formation of runback ice. Tests were conducted at 56m/s and 110m/s (with an aluminium NACA0012 aerofoil in an IWT). The tested system achieved complete anti icing while reducing energy consumption by up to 60%.	Difficutl to manufacture into the CFRP laminate without compromising structural integrity	Leading edge of the wing and rotor blades.	Used to show complaince in the 45 min hold test. Additionally, aids in satisfying general VTOL power reserve regulations.	https://doi.org/10.1109/CoMEA66280.2025.11241799
Slippery Polyurethane	A 2 part solvent borne polyurethane elastomer formulated with surface modifying polymers (SMP-1 to SMP-3). The coating can easily be sprayed and is cured within 7 days at ambient conditions.	The surface modifying polymers simultanously migrate to the surface during curing, enriching the surface with high fluorine content (in the range of 13-29 at.%). The SPU coatings showed an IAS in the range of 220-400KPa. Coating passed ASTM G76 and G73 standards tests, demonstrating excellent erosion resistant against both high speed solid particles and water droplets.	Unlike fragile air pocket coatings, the SPU survives water droplet erosions testing at impingement velocities of 175m/s for over 20 minutes, without experiencing material loss.	Ice adhesion strength (220kPa) is significantly greater than other coatings such as SLIPS(<10kPa), relying primarily on aerodynamic wind to push the ice off.	Areas experiencing high speed rain, where erosion resistance is key.	Can be used inconjuction with an electro thermal heater. Certification demands consistent performance of the IPS across the entire flight envelope, this coating will not degrade and thus can be used to demonstrate complaince for that requirement.	https://doi.org/10.3390/app12199589
odified polyurethane Top Coats	Three component polyurethane topcoat (comericially sold as Aviox 77702) modified by dispersing 3 wt% nonosilica and 5 wt% POSS14 (sphereosilicate) directly into the paint before spraying.	The nanosilica creates a huerarchial surface roughness, while the POSS14 acts as a surface energy modifier, transitioning hydrophilic paint into a highly hydrophobic state with a sliding angle of 30°.	Tested on the NACA0012, the results show this coating being optimised for Glaze Icing conditions. A 65% reduction in accreted glaze ice mass was observed at temperatures of -5℃.	Effectiveness of the coating decreases in colder conditions such as freezing Rime ice (-15℃), droplets freeze upon impact without experieincing any runback, this reduces ice shedding efficiency.	Used on control sources, were glaze ice runback is the primary threat.	Glaze horns can't form as a result of its effectiveness in glaze ice conditions, reducing aerodynamic degredations in drag and lift. This helps in keeping efficiency losses low.	https://doi.org/10.3390/ma14195687

Table 20: [APPENDIX C] Passive Coating Matrix Part 2

ɔating Strategy	Configuration/Method	Mechanism	Strengths	Weaknesses	Possible Applications	Certification Alignment	Source of Study
ırt Temperature sponsive SLIPS	A microcone composite micropillar array (MCMA) fabricated used femtolasers and a soft transfer process. The polydimethylsiloxane (PDMS) matrix is infused with a lubricating mixture of liquid paraffin and silicone oil.	Polymer network that autonomously secretes lubricant oil when exposed to freezing conditions, this creates a slippery liquid interface resulting in low ice adhesion strength. The polymer corsslinks relax and re absorb the oil upon return to warmer conditions.	Low ice adhesion (2.51kPa). Unlike conventional SLIPS that lose their effectiveness at high airspeeds (due to oil being blown off), this coating circulates the oil internally, minimising the effect of water/dust contamination and wind shear. This coating maintains a ice adhesion value <20 kPa even after 50 severe icing/deicing cycles. Forst Formation is also delayed by up to 5 hours.	Application of coating is complex, may not be economical for use on the entire aircraft wing.	Can be used on critical sensors (ice detection sensors) or any control surfaces requiring long term passive ice protection.	The self replenishing oil mechanism minimises the chance of anti icing performance suddenly decaying, provides certification authority proof of reliability.	https://doi.org/10.1016/j.cej.2025.160689
th carbonised spent ɔffee grounds	This eco friendly strategy embeds thermally carbonised spent coffee grounds (5wt%) into a polyurethane resin, producing a highly porous carbon structure. Fun Fact: the research team collected the coffee grounds from their local starbucks which is crazy.	The highly porous CSCG drastically minimises the polar surface energy (close to 0), achieving a Cassie-Baxter State. This forces ice to nucleate in loosly fragemented domains rather than being concentrated.	The paper refers the Adhesion Strength as "Work of Adhesion", this is the fundamental energy required to separate two phases that are in physical contact (the water/ice with the aircraft coating). The CSCG CFRP coating reduced the work of adhesion to 55.81 mJ/m^2 from 88.68mJ/m^2(for unmodified polyurethane).	The material experiences increased brittleness, the tensile strain of the coating experiences a significant decrease. Additionally, large coffee ground particles (>300micrometre) can lower overall mechanical performance .	Can be used on rigid structural components	Eco Friendly?	https://doi.org/10.3390/ma18194533
ıodified Carbon ıbes and Graphene Coating	An epoxy matrix made up of Carbon Nanotubes and Graphense. Rather than using fluorosilanes (like most epoxy coatings), the surface is trated in a vacuum oven. This allows the carbon to naturally absorb hydrocarbons, achieving low surface energy	non colvant (pi-pi) stacking interactions between the CNT's and Graphene create robust hierarchical agglomerations that trap air and repel water. While the structure passively repels water and delays freezing, the carbon nanomaterials also leverage photothermal conversion. (Essentially works using solar energy, converting it into heat to melt the ice).	Thhis coating survived 1000# sandpaper abrasion under a 200g load with minimal contact angle degradation. Delayed droplet freezing by 7 times, and in simulated freezing rain, the coating delayed ice formation by 5 times compared to an Aluminium substrate. Ice melted within 223s due to the photothermal properties of the coating.	The photothermal deicing effect isn't effective during night flights or inside dense cloud cover. Moreover, during repeated anti icing and de icing cycles, the coating was found to be vulnerable to mechanical degradation.	Can be utilised during daytime operation or in climates with less dense cloud cover.	Since the coating also provides heating, this can be recognised as a secondary anti icing system, acting as a supplementary thermal source that requires zero electric draw.	https://doi.org/10.1016/j.surfcoat.2025.132919
lanoparticle Deicing Fluid	Conventional ehtylene glycol-based deicing fluid enhanced by incorportating 3wt% silica nano particles directly into the liquid	The hot glycal based fluid melts the accumulated ice, as the fluid evaporates, silica nanoparticles are left behind. These nanoparticles fill in surface defects and temporarily grant a superhydrophobic Cassie-Baxter state on the aircraft's topcoat.	When tested over 5 icing and deicing cycles, the icing coverage rate reduced to 0.08 (from 0.23). Vertical ice growth decreased from 0.07 to 0.04 over 5 cycles. Lastly, an improved deicing rate was achieved, 14.7mm^2/s , whereas the standard unenhanced fluid achieved 10.1mm^2/s.	Strictly a temporary, ground operational fix. Fluid washes away, a permanent in flight passive coating is still required.	Can be implemented during ground operations for winter maintainence.	Certification requires analysis on ice growth between usage cycles. Typical ground fluids chemically degrade the topcoat which actively promotes faster ice growth on subsequent flights. This fluid ensures the ice growth rate remains stable and predictable	https://doi.org/10.1016/j.porgcoat.2025.109763